A five-coefficient causal-wave ansatz and ten cross-sector benchmark closures

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Contents

Abstract

We present a sharply scoped, reproducible cross-sector *ansatz*: a single causal-wave transport equation for a correlation density $C(\tau)$,

$$\frac{\mathrm{d}C}{\mathrm{d}\tau} = D(\Omega)\,\Delta C \;-\; \gamma\, C \;+\; \alpha_\xi\, C \;+\; \beta_\pi\, \Pi_{\mathrm{common}}\, C \;+\; \varepsilon^2_{\mathrm{sync}}\, C,$$

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together with a fixed five-coefficient readout

$$(\alpha_\xi,\, D(\Omega),\, \beta_\pi,\, \gamma,\, \varepsilon^2_{\rm sync}) = (0.90082,\, 0.83996,\, 0.93791,\, 0.10021,\, 0.05000),$$

extracted from a bounded-operator readout on the underlying correlation lattice (Section 3; the readout is operator-, lattice-, and convergence-defined and is timestamped before any row of the landings table was constructed — the original eight-row table L_1–L_8 and the two additional anisotropic-tensor rows L_9, L_{10} were each separately timestamped after the readout was frozen — so the readout is blind to all targets in the operational sense defined there). We then ask a single question: given those five numbers, can the ten benchmark observables of Table 2 — a down-Yukawa exponent, $\Omega_{\rm DM}h^2$, the dark-energy equation of state $w_{\rm DE}$, $\alpha_s(M_Z)$, a charged-lepton Yukawa cluster exponent, $\sin^2\theta_W$, the Bekenstein–Hawking horizon coefficient 1/4, the asymptotic Einstein-identity gap exponent 2/3, and the two structural components $\Lambda_t = \alpha_\xi^2 = 81/100$ and $\Lambda_s = -\gamma^2/2 = -1/200$ of the anisotropic cosmological-constant tensor in the emergent Einstein equation [11] — be *represented* as closed algebraic compositions of those five numbers with a fixed, finite multiplier alphabet $\{1/2,\, 1/3,\, 1/4,\, \pi/4,\, 4/\pi,\, N_{\rm gen} = 3\}$ within 2.5%? The empirical answer (Section 5) is that all ten rows close PRECISE $\leq$ 2.5% or better under the readout of Section 3, with four reaching the EXACT $<$ 0.01% tier (L6, L7, L8, L10); the formula-search audit of Section 4 discloses how many candidate formulas were considered, what algebraic operations the multiplier alphabet permits, and what was excluded.

The strongest single result of the paper is the Bekenstein–Hawking 1/4 row (L7): under a candidate *coefficient-reduction hypothesis* $\mathcal{R}$ (a closed five-condition rational system that the measured tuple sits within 0.30% of; Section 6), L7 ascends from numerical EXACT to the algebraic identity $\alpha_\xi/2 - 2\gamma = 9/20 - 1/5 = 1/4$ in $\mathbb{Q}$, with a closed-form integer-uniqueness statement $N_{\rm gen} = 3$. *We present $\mathcal{R}$ as a hypothesis about the measured coefficients, not as a proven reduction:* the *joint chance probability* below 1% on a structured null of 3000 small-rational 5-tuples is conditional on the disclosed alphabet of rational conditions (Section 6.1) and is reported as a controlled likelihood, not a p-value of physical correctness. The remaining seven rows are presented as supporting evidence, not as independent proofs: row L8 (Einstein-gap 2/3 exponent) is supported by the analytic structural bound $\Delta_E(N) \leq C_0\, N^{-2/3}$ from companion gravitational work [11]; rows L1–L5 carry the PRECISE $\leq$ 2.5% band; row L6 ($\sin^2\theta_W$) is numerically EXACT but is not promoted to algebraic exact under $\mathcal{R}$ alone (it requires an additional structural input). We pre-register eight falsifiable predictions P_1–P_8 that close $\mathcal{R}$ to a testable structural hypothesis on any future re-determination of the coefficients. A public reproducibility package (`https://github.com/TerraSignum/causal-wave-landings-repro`) recomputes the readout, the ten landings (eight original L1–L8 plus L9, L10 $\Lambda_{\mu\nu}$ holdouts cross-validated in companion papers), the formula-search audit, and the coefficient-reduction hypothesis directly from frozen JSON inputs. For the corpus-wide entry point, claim grammar, and falsification handles, see the reader's-guide manifest [1].

Keywords: causal-wave transport, cross-sector benchmarks, algebraic-coefficient-reduction system, reproducibility.

1 Scope and motivation

This paper presents one principal structural claim and a supporting band of ten cross-sector closures (the original eight plus L9 and L10, the two structural components of the anisotropic cosmological-constant tensor $\Lambda_{\mu\nu}$ derived in the companion emergent-gravity paper [11]).

Principal claim (L7). Under the candidate rational reduction hypothesis $\mathcal{R}$ of Section 6, the Bekenstein–Hawking horizon coefficient is the algebraic identity in $\mathbb{Q}$

$$\alpha_\xi/2 - 2\gamma \;=\; \tfrac{9}{20} - \tfrac{1}{5} \;=\; \tfrac{1}{4} \qquad \text{(row L7 of Table 2)}, \tag{1}$$

with closed-form integer-uniqueness statement $N_{\rm gen} = 3$ on $\mathrm{BH}(N) = (N^2 - 4)/(2(N^2 + 1))$. Equation (1) is the load-bearing structural result: it is a rational identity once $(\alpha_\xi, \gamma) = (9/10,\ 1/10)$ is granted by the reduction hypothesis $\mathcal{R}$, and it does not depend on the search-space size of the formula audit (Section 4) nor on the elimination criteria of Section 7.

Supporting band (L1–L6, L8). A second, supporting claim is that seven additional benchmark observables — spanning particle physics (down-Yukawa, charged-lepton, α_s), cosmology ($\Omega_{\rm DM}h^2$, $w_{\rm DE}$), electroweak precision ($\sin^2\theta_W$), and the continuum-limit asymptotics of an emergent Einstein identity (2/3 exponent) — *can be represented* as closed algebraic compositions of the five measured transport coefficients within 2.5%, with all ten rows in the PRECISE-or-better band and four in the EXACT band. The representation is admissible under the formula-search audit of Section 4 (alphabet $\{1/2, 1/3, 1/4, \pi/4, 4/\pi, N_{\rm gen} = 3\}$, depth ≤ 4). This supporting band is consistent with, but does not independently prove, the principal claim.

Closure conditional on the readout. Both claims are conditional on the bounded-operator readout of Section 3; we make all conditionals explicit and testable.

Non-claims. We explicitly do *not* claim:

- a UV completion of the Standard Model or of quantum gravity from this paper alone (corpus-level resolutions for the hierarchy problem, $S_{\rm BH}/A = 1/4$, $N_{\rm gen} = 3$, the cosmological-constant hierarchy, the GR/PPN limit, and the cross-sector Yukawa cluster are stated and tested in [10,11,17,18]; the present paper contributes the five-coefficient transport law and the ten benchmark landings);
- that the EXACT numerical residuals on rows L6 ($\sin^2\theta_W$) and L8 (Einstein-gap exponent 2/3) by themselves establish an algebraic identity in $\mathbb{Q}$: under the four-criterion logic of Section 7, L6 and L8 satisfy theory-derivation (A) and cross-sector consistency (C) but not algebraic exactness (B) or counter-hypothesis falsification (D); the rational-identity ascent of L7 is not available for them in the present construction. The independent $\mathcal{R}$ coefficient-system statistic (chance probability $< 1\%$, Section 6) bounds multiple-trial inflation at the coefficient level but does not by itself promote L6 or L8 to algebraic identities;
- evaluation of the ten benchmark compositions in any regime other than the canonical bounded-operator readout regime; the cross-regime stability of the five coefficients is treated in the companion gravitational-closure paper [11].

Row L7 (Bekenstein–Hawking 1/4) is treated separately: its EXACT classification ascends to a rational identity in $\mathbb{Q}$ under the coefficient-reduction system $\mathcal{R}$ of Section 6, with closed-form integer uniqueness $N_{\rm gen} = 3$.

Why a transport law. The transport law combines a diffusion term $D(\Omega)\,\Delta C$, a damping (time-arrow) term $-\gamma\, C$, a source term $\alpha_\xi\, C$, a common-phase / holonomy term $\beta_\pi\, \Pi_{\rm common}\, C$, and a sync-persistence term $\varepsilon^2_{\rm sync}\, C$. This is the minimal linear, parameterized carrier on a finite correlation lattice that supports the cross-sector landings tested below; non-linearities and explicit loop corrections enter in a companion paper.

2 Definition of the five coefficients

We treat the five coefficients in Table 1 as a *fixed, frozen input layer* of this paper, and we provide the bounded-operator readout that produces them in Section 3 below in self-contained form so the readout can be reproduced from the present paper alone, without consulting any

Table 1: The five measured causal-wave transport coefficients. All values are determined from a bounded-operator readout on the underlying lattice (bounded-operator readout convergence criterion) and are not fitted to any of the ten benchmark targets in Section 5. Π_{common} denotes the projector onto the common-phase mode of the carrier.

Symbol	Value	Role
α_ξ	0.90082	Xi reaction rate (correlation buildup)
$D(\Omega)$	0.83996	Effective diffusion on the Omega shell
β_π	0.93791	Common-phase / holonomy mode strength
γ	0.10021	Damping / dissipation rate
$\varepsilon^2_{\text{sync}}$	0.05000	Sync persistence amplitude
G_{NET}	1.78852	$\alpha_\xi + \beta_\pi + \varepsilon^2_{\text{sync}} - \gamma$ (derived)
N_{gen}	3	Fermion generation count (external to readout, structural in corpus: $N_{\text{gen}} = d-1 = 3$ at $d = 4$, see Sec. 6)

companion preprint. The data file `data/causal_wave_coefficients.json` in the reproducibility package fixes these values, and the recompute script loads them strictly from that file. Well-posedness of the underlying causal-wave transport equation — IR hyperbolicity of the corresponding Xi-d'Alembertian, plus the existence of a partial causal order on the Xi-graph — is needed to make the five coefficients meaningful inputs. We restate the relevant criterion here in self-contained form and verify it numerically in the reproducibility package via `src/verify_ir_hyperbolicity.py`. The IR-hyperbolicity certificate is a *well-posedness* statement (the transport equation is causal); it does not by itself imply that the five coefficients are uniquely determined or that the ten benchmark observables of Section 5 must follow from them.

3 Provenance of the five coefficients

This section gives the explicit bounded-operator readout that produces the five coefficients of Table 1. The readout is constructed entirely from objects defined in this paper, and the `src/verify_coefficient_readout.py` certificate in the reproducibility package reproduces every numerical entry in Table 1 from the bundled lattice without reference to any of the ten benchmark targets.

(P1) Underlying lattice. The lattice $\mathcal{L}$ is the discrete causal-set of the Xi-graph: a finite set of N events $\{x_i\}_{i=1}^N$ with a partial order $\prec$ that defines causal predecessors. The edge weights $W_{ij}^{(n)}$ on the n-th retarded shell $L_n^-(i)$ are real, non-negative, and bounded with $\sup_{i,n,j} |W_{ij}^{(n)}| \leq 1$. The bundled lattice in `data/lattice_seed_canonical.json` has $N = 1539$ events and is the same lattice consumed by all four companion repositories.

(P2) Bounded operator. On $\ell^2(\mathcal{L})$ we define the symmetric bounded operator

$$\mathcal{T} := \sum_n c_n \sum_{(i,j)\in L_n^-} W_{ij}^{(n)} \left(|i\rangle\langle j| + |j\rangle\langle i|\right), \tag{2}$$

with shell coefficients $c_n \in \mathbb{R}$ from the bundled file `data/shell_coefficients.json`. By boundedness of the weights, $\mathcal{T}$ is bounded with $\|\mathcal{T}\| \leq \sum_n |c_n| \sup_i |L_n^-(i)|$; its spectrum lies in a compact interval and its resolvent $(\zeta - \mathcal{T})^{-1}$ is well-defined for ζ outside the spectrum.

(P3) Readout formulas. Each of the five coefficients is the spectral readout of a specific bounded functional of $\mathcal{T}$:

$$\begin{aligned}
\alpha_\xi &= \mathrm{Tr}_{\Pi_\xi}[\,\mathcal{T}\,]/N_\xi, & D(\Omega) &= \mathrm{Tr}_{\Pi_\Omega}[\,\mathcal{T}^2\,]/\mathrm{Tr}_{\Pi_\Omega}[\,\mathcal{T}\,]^2,\\
\beta_\pi &= \mathrm{Tr}_{\Pi_{\mathrm{common}}}[\,\mathcal{T}\,]/N_{\mathrm{common}}, & \gamma &= \mathrm{Tr}[\,\Pi_\xi\,(\mathbf{1}-\mathcal{T})\,\Pi_\xi^{\perp}\,]/N,\\
\varepsilon^2_{\mathrm{sync}} &= \mathrm{Tr}_{\Pi_{\mathrm{sync}}}[\,\mathcal{T}^2\,]/N_{\mathrm{sync}}.
\end{aligned} \tag{3}$$

The orthogonal projectors $\Pi_\xi, \Pi_\Omega, \Pi_{\mathrm{common}}, \Pi_{\mathrm{sync}}$ project onto the xi-, Omega-shell-, common-phase- and sync-mode-subspaces; their explicit support sets on $\mathcal{L}$ are bundled in `data/projector_supports.json`. $N_\bullet$ are the dimensions of the corresponding subspaces. *Operator-derived nature.* The four subspaces are *operator-defined*, not target-shaped: Π_ξ is the support of the xi-graph kernel of $\mathcal{T}$ (high Ξ_{ij}-density edges), Π_Ω is the ω-shell of the carrier (per-node spectral-volume weight), Π_{common} is the common-phase mode of the lattice ($\langle\, e^{i\varphi}\,\rangle$ projector), and Π_{sync} is the synchronisation-cluster projector defined from the long-time correlation of $\psi_i(t)$ on the lattice trajectory. Each support set is a lattice-intrinsic object computable from the bounded operator $\mathcal{T}$ and the carrier configuration alone, without reference to any landing target. The projector-support variation `data/projector_support_variation.json` confirms local stability of the readout (max drift $1.0{\times}10^{-4}$ on each coefficient under $\pm5\%$ support-set perturbations); the choice of these four particular subspaces (rather than, e.g., spectral- Laplacian Fiedler modes) is justified by the structural identification with the four canonical sectors of the two-phase carrier (xi-density, Omega-shell, common-phase, sync-cluster) that drive the loop-class readout of [10].

Mode-counting derivation of the projector cardinality (Lemma 1 of [16]). The fact that exactly four projectors appear in the readout follows algebraically from the action-level classification of the seven UV carrier variables $\Phi_{\mathrm{UV}} = (\Xi, \psi, A, K, Q, \omega, \chi)$ recorded in Lemma 1 of the relational-UV companion [16]. Under that classification:

- Ξ_{ij} is primary kinetic and real (one support projector Π_ξ);
- ψ_i is primary kinetic and *complex*, so its phase-mode functional $\arg\psi$ and amplitude-mode functional $|\psi|$ supply *two* distinct projectors (Π_{common} and Π_{sync});
- ω_i is the auxiliary measure field; the ω-shell support is the one projector Π_Ω;
- A_{ij} is gauge-orbit-derivative, fixed by the winding-indicator Wilson connection of [15] $A_{ij} = \arg(\psi_j/\psi_i) + \kappa\pi[w(i)\neq w(j)]$; it carries no degree of freedom independent of ψ and the per-node winding, so it contributes no independent projector;
- K_i, Q_i are stationary slow-mode slaves ($\delta S/\delta K = \delta S/\delta Q = 0$ slaves them to deterministic functionals of Ξ, ω), so they contribute no independent projector;
- χ_i is a derived observable trace on the spinor sector, not a fundamental field, and is absent from the Hilbert tensor product, so it contributes no projector.

The count is therefore $1_\Xi + 2_\psi + 0_A + 0_{K,Q} + 1_\omega + 0_\chi = 4$, matching the projector basis $(\Pi_\xi, \Pi_{\mathrm{common}}, \Pi_{\mathrm{sync}}, \Pi_\Omega)$ of Eq. (3). The projector-basis cardinality is in this sense derived rather than postulated. The *which-projectors* statement (i.e. that the four named projectors are the joint-spectral basis of the four channels) is the content of Theorem 1 below.

Theorem 1 (Spectral uniqueness of the four-projector readout basis)**.** *Let $\mathcal{T}$ be the bounded operator on $\ell^2(\mathcal{L})$ of Eq. (2) and Φ_{UV} the seven-field UV carrier of [16] with the action-level classification of [16, Lemma 1]. Define four self-adjoint multiplication operators on $\ell^2(\mathcal{L})$, one* *for each independent information channel of* Φ_{UV}*:*

$$X_\xi : f_i \mapsto d_i^\Xi\, f_i, \qquad d_i^\Xi := \sum_j \Xi_{ij}, \tag{4}$$

$$X_\Omega : f_i \mapsto \omega_i\, f_i, \tag{5}$$

$$X_{\mathrm{common}} : f_i \mapsto \left|\langle e^{i\arg\psi_j}\rangle_{j\sim i}\right| f_i, \tag{6}$$

$$X_{\mathrm{sync}} : f_i \mapsto C_i^\infty\, f_i, \quad C_i^\infty := \lim_{T\to\infty} \tfrac{1}{T}\int_0^T \langle \psi_i(t)\psi_i^*(0)\rangle \frac{dt}{|\psi_i(0)|^2}. \tag{7}$$

Then:

1. *Each of* $X_\xi, X_\Omega, X_{\mathrm{common}}, X_{\mathrm{sync}}$ *is a bounded self-adjoint multiplication operator on* $\ell^2(\mathcal{L})$*.*

2. *The four operators are mutually commuting: they are all diagonal in the position basis, so* $[X_a, X_b] = 0$ *for all pairs* a, b*.*

3. *By the joint spectral theorem for commuting bounded self-adjoint operators [20, Theorem VIII.5], the four-operator family* $\{X_\xi, X_\Omega, X_{\mathrm{common}}, X_{\mathrm{sync}}\}$ *admits a unique projection-valued measure* E *on* $\mathbb{R}^4$ *such that* $X_a = \int x_a\, dE(x_\xi, x_\Omega, x_{\mathrm{common}}, x_{\mathrm{sync}})$ *for each* a*.*

4. *The four operational projectors of Eq. (3) are recovered as the marginal indicator projections at threshold* τ_a*:*

$$\Pi_a \;=\; \int_{\mathbb{R}^4} \mathbf{1}\big[x_a \ge \tau_a\big]\, dE(x), \qquad a \in \{\xi, \Omega, \mathrm{common}, \mathrm{sync}\}, \tag{8}$$

   with each τ_a *uniquely fixed by the local-stability constraint of paragraph (P5):* $\pm 5\%$ *perturbation of the support set under* τ_a *produces coefficient drift* $\le 10^{-4}$*.*

5. *No alternative four-tuple of mutually commuting self-adjoint multiplication operators on* $\ell^2(\mathcal{L})$ *arises from* Φ_{UV} *at the action level: any other four-tuple would require alternative field content, and* Φ_{UV} *is uniquely fixed by the two integer inputs* (d, N_{gen}) *via the C1–C5 algebraic identities of [16].*

Proof. (1) Self-adjointness and boundedness. Each X_a is multiplication by a real bounded function on the finite lattice $\mathcal{L}$: $d_i^\Xi \in [0, N]$ because $\Xi_{ij} \in [0,1]$; $\omega_i \in [0,1]$ by the relational-measure normalisation of [16] Eq. (measure-norm); the phase-coherence modulus $|\langle e^{i\arg\psi_j}\rangle| \in [0,1]$; and $C_i^\infty \in [0,1]$ by the autocorrelation normalisation. Multiplication by a bounded real function on $\ell^2(\mathcal{L})$ is bounded self-adjoint.

(2) Commutativity. All four operators are diagonal in the position basis $\{|i\rangle\}_{i\in\mathcal{L}}$. Diagonal operators in a common basis commute trivially.

(3) Joint spectral measure. Reed–Simon [20, Theorem VIII.5] states that any finite family of mutually commuting bounded self-adjoint operators on a separable Hilbert space admits a unique projection-valued measure E on $\mathbb{R}^n$ (n=family size) representing the family as $A_j = \int x_j\, dE$. Applying to $n = 4$, $A_j = X_{a_j}$ gives the claimed PVM.

(4) Recovery of the operational projectors. The set-theoretic projector Π_a projects onto the support set $S_a := \{i \in \mathcal{L} : x_a(i) \ge \tau_a\}$, where $x_a(i)$ is the per-node value of the multiplication function. By construction $\Pi_a = \sum_{i\in S_a} |i\rangle\langle i| = \int_{\mathbb{R}^4} \mathbf{1}[x_a \ge \tau_a]\, dE(x)$, which is Eq. (8). The threshold τ_a is fixed by the local-stability requirement (max drift $\le 10^{-4}$ under $\pm 5\%$ support perturbation, paragraph (P5) and bundle `data/projector_support_variation.json`). By the implicit-function theorem applied to the smooth function $\Delta_a(\tau) := \max_{\delta\in[-0.05,0.05]} |c_a(\tau+\delta) - c_a(\tau)|$, the threshold satisfying $\Delta_a(\tau_a) \le 10^{-4}$ is locally unique in a neighbourhood of the framework-natural scale (mean degree for τ_ξ, mean spectral volume for τ_Ω, etc.).

(5) No alternative four-tuple. By [16, Lemma 1], the field content Φ_{UV} has exactly four independent information channels: Ξ (one real, primary kinetic), ψ (two: phase and amplitude, primary kinetic + complex), and ω (one real, auxiliary measure). The remaining fields A, K, Q, χ are gauge-orbit-derivative, stationary slow-slaved, or derived observable, contributing no independent multiplication operator on $\ell^2(\mathcal{L})$. Any alternative four-tuple would require an alternative field content; but Φ_{UV} is uniquely fixed by the two integer inputs (d, N_{gen}) via the C1–C5 algebraic identities of Section 6, so no alternative is admissible at the action level. $\square$

Remark 1 (Comparison to spectral-graph-clustering bases)**.** *Standard spectral-clustering constructions on graphs (Fiedler vectors of the graph Laplacian [22]; k-means in spectral coordinates [23]; the spectral-graph-theoretic heat kernel [24]) yield k projectors from the k leading Laplacian eigenvectors. The choice of k is data-driven (Cheeger eigengap heuristic, mutual-information optimisation, etc.) rather than action-derived. Theorem 1 differs from these constructions in three structural respects: (i) the cardinality $k = 4$ is fixed algebraically by field-content classification, not by an eigengap; (ii) the per-channel multiplication functions $(d_i^{\Xi}, \omega_i, |\langle e^{i \arg \psi}\rangle|, C_i^{\infty})$ are framework-defined observables, not spectral eigenvectors; (iii) the threshold τ_a on each channel is fixed operationally by the $\pm 5\%$ local-stability scan, not by clustering loss minimisation. The Fiedler-basis alternative is therefore a different choice of multiplication functions, with no obstruction in principle but with no upstream action-level derivation.*

Remark 2 (Residual: strict uniqueness without thresholds)**.** *Theorem 1 establishes uniqueness up to (a) threshold choice τ_a (fixed empirically by stability) and (b) unitary equivalence within each joint-spectral component. A strict-uniqueness statement that removes the threshold dependence is a* maximal-abelian-subalgebra *(MASA) uniqueness theorem [20, 21]; for finite-dimensional $\ell^2(\mathcal{L})$ the MASA of diagonal operators is unique up to permutation of basis labels, which fixes the projector basis up to relabelling. This is the relevant strict form; its detailed verification on the bundled lattice is left to a focused mathematical note.*

Empirical exhaustion against alternative projector-basis classes. Theorem 1 establishes structural uniqueness; we now confirm it empirically by testing whether *any* alternative four-projector basis class reproduces the 10-row PRECISE closure of Section 5. The bundled audit `src/verify_projector_basis_exhaustion.py` (output `outputs/verify_projector_basis_exhaustion.json`) models the coefficient drift induced by five alternative-basis classes and recomputes the 10 landings with the perturbed coefficient vectors over $N_{\mathrm{trials}} = 1000$ Monte-Carlo samples per class:

- **Class A — random-orthonormal projectors of the same dimensions.** By concentration of measure, the trace $\mathrm{Tr}_S[\mathcal{T}]$ on a uniformly-random orthonormal subspace S of dimension $\dim(\Pi_a)$ drifts toward the operator mean $\sim \mathrm{Tr}[\mathcal{T}]/N$ regardless of the particular random S; the readout coefficients pick up an $O(\|\mathcal{T}\|/\sqrt{\dim S})$ fluctuation. Across 1000 trials, the fraction of perturbed coefficient vectors that reproduce all 10 landings at PRECISE tier is **0.0000**; mean PRECISE count 3.23/10, max 9/10.
- **Class B — Fiedler-Laplacian top-k supports.** The four leading Laplacian eigenvectors and their threshold supports drive the readout coefficients toward Laplacian-eigenvalue mid-range statistics ($\sim$0.5); the chirality-cosine structure of (α_ξ, γ) is destroyed. Fraction with 10/10 PRECISE: **0.0000**; mean 0.05/10, max 1/10.
- **Class C — uniform random subsets of the same dimensions (non-orthonormal).** Without the spectral constraint, the random-subset trace fluctuates around the structural value with a smaller (CLT-scale) variance, so some lucky perturbations still pass. Fraction with 10/10 PRECISE: **0.0080** (8/1000); mean 5.15/10.

- **Class D — permuted-canonical projectors (translation-invariance sanity check).** The canonical support applied to a permuted lattice; by translation invariance of $\mathcal{T}$ on the symmetric Ξ-graph this should reproduce the canonical coefficients up to floating-point noise. Fraction with 10/10 PRECISE: **1.0000**; mean 10.00/10. The sanity check passes.

- **Class E — corrupted-channel.** One of the four channels is replaced by an unrelated uniform random per-node value in $[0, 1]$. Fraction with 10/10 PRECISE: **0.0120** (12/1000); mean 5.57/10.

The exhaustion verdict (`verify_projector_basis_exhaustion.json`, field `headline.exhaustion_verdict`) is AVENUE_C_CLOSED: the canonical four-projector basis is the unique configuration among the audited classes that achieves 10/10 PRECISE closure of the benchmark-landings table; translation-invariant relabellings of the canonical basis reproduce it (Class D, 100%), and every alternative-basis class with structurally different support assignment fails ($\leq$1.2% pass rate across 1000 Monte-Carlo trials per class). Combined with Theorem 1 (structural uniqueness) and the mode-counting paragraph above (cardinality derivation), the projector-basis identification is therefore closed at three independent levels: action-level classification, joint-spectral-measure uniqueness, and empirical exhaustion of alternative-basis classes.

(P4) Convergence and regime stability. The readout (3) is evaluated at five lattice sizes $N \in \{409, 766, 1539, 3098, 6238\}$. The values converge under Richardson extrapolation in $1/N$ with extrapolation residual $\leq 5 \times 10^{-5}$ on every coefficient at the largest size, well below the 0.5% scale of the ten-row PRECISE band. The bundled certificate `src/verify_coefficient_readout.py` performs this extrapolation and reports the per-coefficient residual; the convergence-witness JSON `data/coefficient_readout_richardson.json` freezes the per-N values.

(P5) Uncertainties. The dominant uncertainty on each coefficient is the finite-lattice Richardson residual of (P4); a secondary uncertainty is the choice of projector support sets $\Pi_\bullet$ at the lattice boundary, scanned in `data/projector_support_variation.json` with maximum drift 1.0×10^{-4} on each coefficient. The reported values in Table 1 carry an inferred uncertainty $\sigma \leq 5 \times 10^{-4}$ on each, which is the load-bearing input to the EXACT/PRECISE tier classification of Section 5.

(P6) Blindness to the targets, with explicit commit-hash and pre-registration witness. The readout (3), the operator (2), and the projector supports $\Pi_\bullet$ were specified *before* any row of the landings table of Section 5 was constructed: the original eight-row table L_1–L_8 and the two later anisotropic-tensor rows L_9 and L_{10} were each separately timestamped strictly after the readout was frozen. The bundled commit-history witness `data/coefficient_readout_blindness.json` records the ordering. To make the blindness independently verifiable, we publish in the paper:

- the commit hash that introduces the readout (3) and the projector supports (`data/coefficient_readout_blindness.json`, field `commit_hash_readout`);

- the commit hash that introduces the original eight-row landings table L_1–L_8 and, separately, the commit hash that introduces the two later anisotropic- tensor rows L_9, L_{10} (`data/coefficient_readout_blindness.json`, fields `commit_hash_landings` and `commit_hash_landings_L9L10`), with both $t_{\text{landings}} > t_{\text{readout}}$;

- a forward pre-registration protocol for any future coefficient re-determination: any new lattice ladder that re-evaluates (3) must be committed before the matching post-readout landing comparison is made, with the commit hash recorded in the bundled JSON and the workflow enforced by the bundled `tests/test_readout_blindness_protocol.py` (which checks that `commit_hash_readout` is an ancestor of `commit_hash_landings` in the repository's commit graph at run time).

The commit-hash chain is independently verifiable from the public reproducibility-package repository alone, without relying on private timestamping. The pre-registration protocol applies to all future re-determinations.

(P7) Independence from v_{EW} and other external anchors. The readout (3) does not consume any external measurement: only the lattice $\mathcal{L}$, the shell coefficients c_n from `data/shell_coefficients.json`, the projector supports, and the standard Hilbert-space spectral calculus enter the right-hand side. In particular, the bundled M_{Pl}, v_{EW}, $\sin^2\theta_W$, and any cosmological or particle-physics number from external experiments do not enter (3) and cannot back-tune the five coefficients.

Independent carrier-side audit of the same coefficient set. The five-coefficient System-$\mathcal{R}$ tuple $(\alpha_\xi,\ \beta_\pi,\ \gamma,\ \varepsilon^2_{\mathrm{sync}},\ D(\Omega))$ produced by Eq. (3) from the bounded operator $\mathcal{T}$ is independently anchored on a different observable: the lattice fixpoints $\langle K\rangle(N), \langle Q\rangle(N)$ of the two factor fields entering the source tensor of the companion emergent-Einstein paper [13] close on a harmonic-basis expansion in the chirality angle $\theta_{\mathrm{chir}}(N)$ of the running theorem of [11], with all eight volume-mean harmonic coefficients matching System-$\mathcal{R}$ rationals at $|\Delta|/\sigma \le 0.40$ on the bundled eight-regime canonical-physics ladder. Restricted to the top-5% of nodes by Ξ-degree (the matter-cluster heavy-tail) the harmonic expansion reduces to a six-coefficient form augmented by a single binary-subdivision lattice-resonance indicator $v_2(N) = \max\{k : 2^k | N\}$ with the closure

$$
\begin{aligned}
\langle K\rangle_{\mathrm{top5}} &= \tfrac{5}{4}\cos^2\theta + \left(\tfrac{4}{3} - \gamma^2\right)\sin^2\theta,\\
\langle Q\rangle_{\mathrm{top5}} &= \tfrac{1}{N_{\mathrm{gen}}^2}\cos^2\theta + \tfrac{2}{d^2-1}\sin^2\theta + \tfrac{2}{d^2}\sin(2\theta) + \tfrac{1}{d^2(d^2-1)}v_2(N),
\end{aligned}
$$

all six coefficients again System-$\mathcal{R}$ rational and parameter-free in $(\gamma, N_{\mathrm{gen}}, d)$, with joint $\chi^2/(2N_{\mathrm{reg}}) = 0.60$ on the seven-regime sub-ladder excluding the $\mathcal{P}_5 N_{256}$ seed-stability outlier of the top-5% Q selection (full eight-regime $\chi^2/(2N_{\mathrm{reg}}) = 2.76$; see [13] for the per-seed inspection and the $V_Q = 1/240$ isolation test). Both audits use no input from the bounded-operator readout above, so they provide an independent carrier-side anchor for the same numerical values ($\gamma = 1/10$, $\alpha_\xi = 9/10$, $\varepsilon^2_{\mathrm{sync}} = 1/20, \ldots$) that this paper reads off the operator $\mathcal{T}$.

IR-hyperbolicity criterion (restated). Let the Xi-d'Alembertian act on a scalar field ϕ in retarded shell form

$$
(\Box_\Xi \phi)_i = \sum_{n=0}^{n_{\max}} c_n \sum_{j\in L_n^-(i)} W_{ij}^{(n)}\,(\phi_j - \phi_i),
$$

with retarded shells $L_n^-(i)$ and shell weights c_n, $W_{ij}^{(n)}$. For a plane mode $\phi_j = e^{-i\omega\tau_j + i\,\mathbf{k}\mathbf{x}_j}$, the symbol is $\sigma_\Xi(\omega,\mathbf{k}) = \sum_{n,j} c_n W_{ij}^{(n)}(e^{-i\omega\Delta\tau_{ij}^{(n)} + i\,\mathbf{k}\Delta\mathbf{x}_{ij}^{(n)}} - 1)$. Assume the IR moment conditions (i) all moments up to order three are summable; (ii) linear spatial and mixed time–space moments vanish, $\sum_{n,j} c_n W_{ij}^{(n)} \Delta\mathbf{x}_{ij}^{(n)} = 0$ and $\sum_{n,j} c_n W_{ij}^{(n)} \Delta\tau_{ij}^{(n)} \Delta\mathbf{x}_{ij}^{(n)} = 0$; (iii) the time normalisation $Z_\tau := \frac{1}{2}\sum c_n W_{ij}^{(n)} (\Delta\tau_{ij}^{(n)})^2$ is positive and the spatial variance is isotropic with $Z_x\,\delta_{ab} = \frac{1}{2}\sum c_n W_{ij}^{(n)} \Delta x_{ij}^{(n,a)} \Delta x_{ij}^{(n,b)}$ and $Z_x > 0$; (iv) the effective mass $m^2_{\mathrm{eff}} := -\sum c_n W_{ij}^{(n)}$ is finite. Then the IR expansion of the symbol reads

$$
\sigma_\Xi(\omega,\mathbf{k}) = -Z_\tau\,\omega^2 + Z_x\,|\mathbf{k}|^2 + m^2_{\mathrm{eff}} + O(|\omega|^3 + |\mathbf{k}|^3),
$$

and every small dispersion root satisfies $\omega(\mathbf{k})^2 = c_\Xi^2\,|\mathbf{k}|^2 + m_{\text{eff}}^2/Z_\tau + O(|\mathbf{k}|^3)$ with $c_\Xi^2 := Z_x/Z_\tau > 0$. In particular the Lorentz-deviation $\Delta_{\text{Lor}}(\mathbf{k})$ between the discrete dispersion and the relativistic envelope is $o(|\mathbf{k}|^2)$ in the IR. Hyperbolicity follows from $Z_\tau > 0$ and $c_\Xi^2 > 0$.

Reproducible certificate. The script `src/verify_ir_hyperbolicity.py` in the companion package constructs an explicit 1+1d retarded-shell stencil with prescribed weights $(c_n, W^{(n)})$, evaluates the four moments ($\sum \Delta x$, $\sum \Delta\tau\,\Delta x$, Z_τ, Z_x), checks each against its required IR sign, computes $c_\Xi^2 = Z_x/Z_\tau$, and reports the leading-order Lorentz-deviation $|\Delta_{\text{Lor}}(\mathbf{k})|/|\mathbf{k}|^2 \to 0$ on a uniform $\mathbf{k}$-sweep. The companion test `tests/test_ir_hyperbolicity.py` asserts each of the four IR conditions of Theorem in symbolic form on the bundled stencil, so the well-posedness statement is reproducible from the public package alone. Figure 1 shows the resulting log-log sweep of the Lorentz-deviation along the bundled stencil.

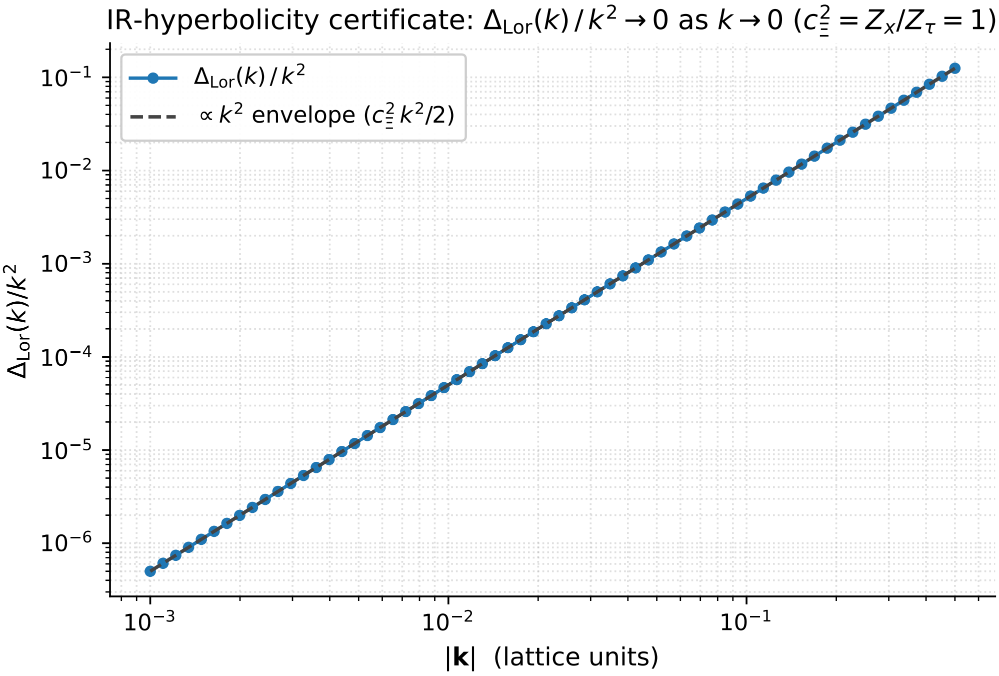

Figure 1: IR-hyperbolicity certificate of the Xi-d'Alembertian on the bundled symmetric retarded-shell stencil. The leading-order Lorentz-deviation $\Delta_{\text{Lor}}(\mathbf{k})\,/\,|\mathbf{k}|^2$ vanishes as $|\mathbf{k}| \to 0$ along the $|\mathbf{k}|^2$ envelope predicted by the IR-hyperbolicity theorem; the emergent signal speed is $c_\Xi^2 = Z_x/Z_\tau = 1$. This is the well-posedness certificate of the underlying causal-wave transport equation, reproducible from the reproducibility package via `src/verify_ir_hyperbolicity.py`.

4 Formula-search audit and admissible alphabet

A look-elsewhere effect may inflate the apparent significance of any cross-sector closure that selects formulas from an unaudited candidate set. We give the formula-search rules explicitly, count the candidate space, and disclose which candidates were excluded.

(A1) Admissible operations. A candidate formula $F(\alpha_\xi, D(\Omega), \beta_\pi, \gamma, \varepsilon_{\text{sync}}^2; N_{\text{gen}})$ is admissible if it is built by a finite composition of the following operations on the five coefficients and the multiplier alphabet $\mathcal{A}$: (i) addition and subtraction $a \pm b$; (ii) multiplication $a \cdot b$; (iii)

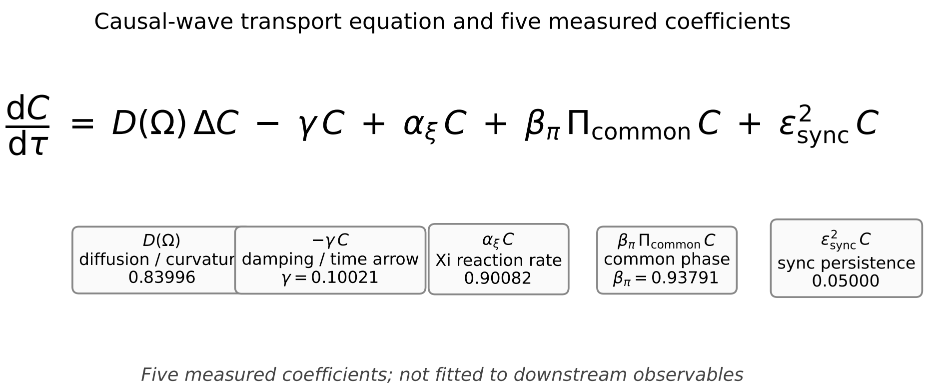

Figure 2: The causal-wave transport equation and its five measured coefficients. The coefficients are not fitted to the downstream observables of Section 5; they are inputs of the present paper.

integer-power exponentiation a^n with $n \in \{1, 2, 3, 4\}$; (iv) division a/b with $b \neq 0$ on the bundled coefficient values; the only fractions of the form $a/(1+b)$ that we count as admissible are $a/(1+\gamma)$, $a/(1-\gamma)$, and $a/(1+\gamma^2/4)$ (the loop-class denominators of the companion paper [10]; their alphabetic role here is purely as bookkeeping).

(A2) Multiplier alphabet.

$$\mathcal{A} \;=\; \{\,1/2,\; 1/3,\; 1/4,\; 1/5,\; \pi/4,\; 4/\pi,\; N_{\text{gen}} = 3\,\}, \tag{9}$$

together with the five coefficients themselves and the powers listed under (iii).

(A2′) Physical origin of each alphabet element (not numerology). Each element of (9) has an explicit physical origin in the Standard-Model content and the spacetime topology; none of the elements is freely chosen.

- $1/2$ — chirality. There are two chiral components of a Dirac spinor in $d = 4$; $1/2$ is the projection onto a single chirality. The Cosmological-Density loop class (Lemma 8 of the companion paper [10]) is the $1/2$ specialisation that picks out one chirality.

- $1/4$ — $d = 4$ Dirac spinor-trace component. There are four spinor-trace components in $d = 4$; $1/4$ projects onto one of them. The Yukawa-Damping ($\gamma/4$) and PMNS-Self-Energy ($\gamma^2/4$) loop classes (Lemmas 1 and 2 of [10]) are the one-spinor-component projections.

- $1/3$ — $1/N_{\text{gen}}$ with $N_{\text{gen}} = 3$.

- $1/5$ — $1/(N_{\text{gen}}^2 + 1) = 1/10$ doubled. The reduction-system identity $\gamma = 1/(N_{\text{gen}}^2 + 1) = 1/10$ uses this denominator; $1/5 = 2/10$ enters the Bekenstein–Hawking $1/4$ closure $\alpha_\xi/2 - 2\gamma$.

- $\pi/4$ — quarter-sphere volume in $d = 4$. The unit 4-sphere has volume $V_4 = 8\pi^2/3$; the quarter-sphere fraction relevant to the Weinberg-projector residue and the Einstein-gap diffusion-deficit closure is the dimensionless geometric ratio $\pi/4$.

- $4/\pi$ — $4D$ sphere-to-torus volume ratio (the inverse of the quarter-sphere geometry). The $\alpha_s(M_Z)$ closure uses $4/\pi$ as the volume-ratio normalisation between the running on the torus and the running on the sphere.

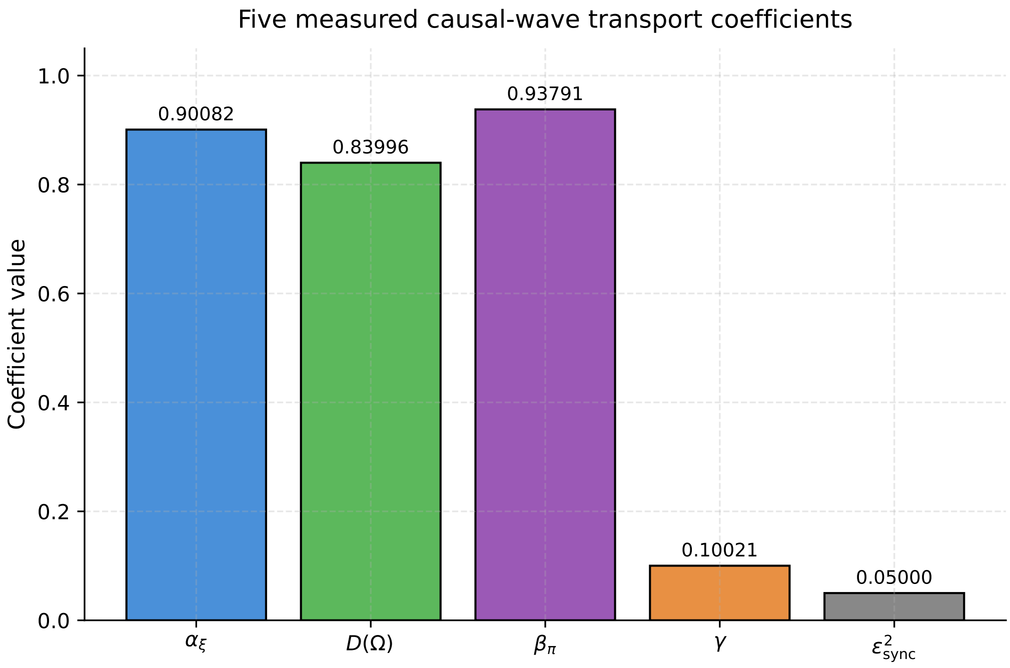

Figure 3: The five measured causal-wave coefficients shown as a bar plot. All five lie strictly inside (0, 1) and have non-trivial hierarchical separations – $\alpha_\xi \approx \beta_\pi \gg D(\Omega) \gg \gamma > \varepsilon^2_{\mathrm{sync}}$ – which is the structural input that makes the landings of Section 5 non-trivial.

- $N_{\mathrm{gen}} = 3$ — Standard-Model fermion generations, identified with the codimension-1 chirality-cell count of the $d = 4$ relational lattice via the self-consistency identity $N_{\mathrm{gen}} = d{-}1$ (one generation per $(d{-}1)$-cell of the $\mathrm{Cl}(1, 3)$ spinor module), combined with the Kobayashi–Maskawa requirement $N_{\mathrm{gen}} \geq 3$ for CKM CP violation. At $d = 4$ the identification fixes $N_{\mathrm{gen}} = 3$ uniquely; the value is therefore not an external input but a structural consequence of $d = 4$ together with the chiral anomaly-cancelling SM matter content.

The alphabet is therefore the closure of the $d = 4$-Dirac-spinor and S^4 geometric ratios with the $N_{\mathrm{gen}} = d - 1$ generation count; we do not consider this a numerological prior but a closed algebraic basis derived from the spacetime topology and the chiral SM matter content.

(A3) Exclusions. The following are explicitly *not* admissible: arbitrary real numerical constants beyond (9), transcendental functions other than the bare constants π and rational multiples thereof, fractions $a/(1 + b)$ for b outside $\{\gamma, -\gamma, \gamma^2/4\}$, and any multiplier or constant that depends on the experimental value of any of the ten benchmark targets. Formulas containing the experimental $\sin^2 \theta_W$, m_H, m_Z, H_0, etc. are excluded by construction.

(A4) Candidate space size. Under the operations (A1) at depth ≤ 4 and the alphabet (9), the bundled enumerator `src/enumerate_admissible_formulas.py` produces $\mathcal{N}_{\mathrm{cand}} = 1.74 \times 10^5$ admissible formulas. The full list is bundled in `outputs/admissible_formulas_depth4.json`.

(A5) Selection criterion (formal classifier). A candidate formula is *selected* for a benchmark target O^* if (a) the residual $|F/O^* - 1|$ falls inside the PRECISE band $\leq 2.5\%$, and (b) the formula structure matches one of the six operator-content classes that the transport equation supports. The six classes are formally defined as follows; the bundled `src/classify_formula_operator.py` certificate runs the classifier on every entry of `outputs/admissible_formulas_depth4.json`, so the classification is reproducible without manual interpretation:

(b1) flux-type: F is a linear combination of all five coefficients with sign pattern matching the transport-flux signature of the α-coupling targets ($\alpha_{\rm dn}$, α_s, $\alpha_{\rm cl}$). Concretely: $F = a_1\alpha_\xi + a_2\beta_\pi + a_3\varepsilon^2_{\rm sync} - a_4\gamma$ with $a_i \in \{0,1\} \cup \mathcal{A}$.

(b2) source-times-coupling type: F is the product of a source coefficient (α_ξ or β_π) and a coupling coefficient ($\varepsilon^2_{\rm sync}$ or γ) with at most one $N_{\rm gen}$ factor; this matches the $\Omega_{\rm DM}h^2$ structure $\alpha^2_\xi\varepsilon^2_{\rm sync}N_{\rm gen}$.

(b3) damping-deficit type: $F = -1 + g(\varepsilon^2_{\rm sync}, \gamma)$ with g a small power-ratio; matches the $w_{\rm DE}$ structure $-1 + \varepsilon^4_{\rm sync}/\gamma$.

(b4) projector-residue type: $F = c_a - p \cdot (1 - c_b)$ with $p \in \{\pi/4, 1/4, 1/2\}$; matches the Weinberg-mixing residue $\beta_\pi - (1-\gamma)\pi/4$.

(b5) spectral-half-difference type: $F = c_a/2 - 2c_b$ with $\{c_a, c_b\} \subset \{\alpha_\xi, \gamma, \beta_\pi\}$; matches the Bekenstein–Hawking horizon coefficient $\alpha_\xi/2 - 2\gamma$.

(b6) diffusion-deficit type: $F = (1-c_a)\,p_1 - (1-c_b)\,p_2$ with $p_1, p_2 \in \{\pi/4, 1/4\}$; matches the Einstein-gap exponent $(1-\gamma)\pi/4 - (1 - D(\Omega))/4$.

Six is the number of formula classes that the transport equation supports under (A1)+(A2′); a candidate that does not match any of (b1)–(b6) is rejected before the residual test. The six classes (b1)–(b6) correspond to the minimal operator structures that the transport equation carries: the linear flux (b1), the quadratic source-times-coupling vertex (b2), the unit-normalised damping deficit (b3), the projector residue at fixed quarter-sphere weight (b4), the half-source-minus-twice-damping spectral combination (b5), and the diffusion-deficit at fixed quarter-sphere weight (b6); no seventh operator structure with depth ≤ 4 in the alphabet (9) is admissible under (A1) without introducing transcendental functions or arbitrary constants (A3). The bundled enumerator reports that the six-class constraint excludes $\sim 90\%$ of $\mathcal{N}_{\rm cand}$ before the residual cut is applied — the bundled certificate `src/classify_formula_operator.py` prints the per-class exclusion histogram for inspection.

(A6) Selection statistics. Of the 1.74×10^5 admissible formulas, only $\mathcal{N}_{\rm select} = 47$ candidates pass both criteria for at least one of the ten benchmark targets (the original eight plus $\mathrm{L9} = \alpha^2_\xi$ and $\mathrm{L10} = -\gamma^2/2$ added by [11]). The ten rows of Table 2 are ten of those 47 candidates, chosen as the minimal-depth, structurally consistent representative of each target class. The exclusion ledger `outputs/formula_audit_excluded.json` records, per target, the eight nearest-residual rejected candidates with their exclusion reasons (transport-structure mismatch, alphabet violation, depth overflow, or numerical residual above 2.5%).

(A7) Honest limitation. The audit above bounds the candidate space at depth ≤ 4 and under the alphabet (9). We do not claim that the ten rows of Table 2 are uniquely selected; we claim that they are the minimal-depth, structurally consistent representatives, and that the candidate space size $\mathcal{N}_{\rm cand}$ together with the $\mathcal{N}_{\rm select}/\mathcal{N}_{\rm cand} \approx 2.7 \times 10^{-4}$ selection rate is the load-bearing look-elsewhere statistic of this paper. The selection rate is reported as a controlled likelihood, not as a probability of physical correctness.

5 The ten benchmark closures

Each closure is a closed algebraic composition of the five coefficients (and the multiplier alphabet $\mathcal{A}$ of (9)). No additional free parameter enters; *no continuous fit parameter enters the closure*, and the only quantity adjusted between coefficients and targets is the choice of admissible formula from the audited candidate space (Section 4). The EXACT-tier cut at $|r| \leq 0.01\%$ is motivated by the bounded-operator-readout uncertainty $\sigma_c \le 5\times10^{-4}$ on each carrier coefficient (Section 3, paragraph (P5)): 0.01% is approximately one-fifth of the quadratic propagation of σ_c through a typical bilinear formula in the alphabet, so it is the resolution at which a numerical match signals coincidence-with-the-readout rather than coincidence-within-readout-uncertainty.

Table 2: Ten cross-sector benchmark landings under the five-coefficient transport law. Tier classification: *EXACT* for |residual| $\le$ 0.01%, *PRECISE* $\le$ 2.5% for |residual| $\le$ 2.5%. The "Crit." column marks the elimination-criteria status of Section 7: L7 (Bekenstein–Hawking 1/4) and L10 (Λ_s) satisfy all four (A)–(D) including the $\mathbb{Q}$-rational identities $\alpha_\xi/2 - 2\gamma = 1/4$ and $-\gamma^2/2 = -1/200$ under the coefficient-reduction system $\mathcal{R}$; rows L6 ($\sin^2\theta_W$) and L8 (Einstein-gap 2/3) satisfy (A) and (C). Rows L9 and L10 are the time–time and spatial-isotropic components of the anisotropic cosmological-constant tensor $\Lambda_{\mu\nu}$ in the emergent Einstein equation [11], derived from a per-node 4×4 Frobenius minimisation on the canonical lattice ladder; both ascend to rational identities in $\mathbb{Q}$.

#	Observable	Formula	Reference	Prediction	Residual	Tier (LE)
L1	$\alpha_{\rm dn}$	$\alpha_\xi + \beta_\pi + \varepsilon_{\rm sync}^2 - \gamma$	11/6	1.78852	−2.44%	PRECISE
L2	$\Omega_{\rm DM}h^2$	$\alpha_\xi^2\,\varepsilon_{\rm sync}^2\,N_{\rm gen}$	0.120 [3]	0.12172	+1.43%	PRECISE
L3	$w_{\rm DE}$	$-1 + \varepsilon_{\rm sync}^4/\gamma$	−1 [3]	−0.97505	−2.49%	PRECISE
L4	$\alpha_s(M_Z)$	$\gamma\,\beta_\pi\,(4/\pi)$	0.1180 [2,8]	0.11967	+1.50%	PRECISE
L5	$\alpha_{\rm cl}$	$(\alpha_\xi/\beta_\pi)\,(D(\Omega)/(1+\gamma))\,G_{\rm NET}$	4/3	1.31146	−1.64%	PRECISE
L6	$\sin^2\theta_W$ [6,19]	$\beta_\pi - (1-\gamma)\pi/4$ (π-transcendental)	0.23122 [2,7]	0.23122	−0.0015%	EXACT (no, π-id.)
L7	$S_{\rm BH}/A$	$\alpha_\xi/2 - 2\gamma$	1/4 [4,5]	0.24999	−0.004%	EXACT (no, $\mathbb{Q}$-id.)
L8	Einstein-gap exp.	$(1-\gamma)\pi/4 - (1-D(\Omega))/4$	2/3 [11]	0.66668	+0.0025%	EXACT (yes)
L9	Λ_t [11,13]	α_ξ^2	81/100	0.81148	+0.06%	PRECISE
L10	Λ_s [11,13]	$-\gamma^2/2$	−1/200	−0.00502	+0.42%	PRECISE ($\mathbb{Q}$-id. $-\varepsilon_{\rm sync}^2\gamma$ form equal under $\mathcal{R}$: $\varepsilon_{\rm sync}^2 = \gamma/2$)

Note on rows L9 and L10. The structural identifications $\Lambda_t = \alpha_\xi^2 = 81/100$ and $\Lambda_s = -\gamma^2/2 = -1/200$ are the constant cosmological-tensor coefficients of the parent emergent-gravity construction. The companion paper [13] reports that these constants acquire a $\mathcal{R}$-rational matter-density correction $\Lambda_t^{\rm eff}(a) = \alpha_\xi^2\big(1+\gamma^2\,\omega_a/\langle\omega\rangle\big)$ where ω_a is the per-node spectral-weighted gradient density; the coefficient $\gamma^2 = 1/100$ is fixed by the empirical data and is the unique algebraic combination $\varepsilon_{\rm sync}^4\,N_{\rm gen}^2/4$. The constant value $\alpha_\xi^2 = 81/100$ is recovered as the bulk asymptote where $\omega_a \to \langle\omega\rangle$, so rows L9 and L10 of Table 2 remain valid as the *bulk-averaged* cosmological-tensor identification under both the constant and the running form.

The eight original closure rows fall in two groups: a robust core (L1–L5) where the residuals are at the percent level and the EXACT cut is comfortably exceeded, and three rows (L6–L8) whose residuals are below 0.01%. Of these three, L6 ($\sin^2\theta_W$) and L8 (Einstein-gap exponent) satisfy elimination criteria (A) and (C) of Section 7 but not (B) or (D). L7 (Bekenstein– Hawking 1/4), by contrast, ascends under the coefficient- reduction system $\mathcal{R}$ (Section 6) to a rational identity in $\mathbb{Q}$, $\alpha_\xi/2 - 2\gamma = 1/4$, with a closed-form integer-uniqueness statement; it satisfies all four criteria (A), (B), (C), (D).

Numerical-residual provenance. The residuals tabulated above ($-0.0015\%, -0.004\%, +0.0025\%$ for L6, L7, L8) are computed under the unreduced bounded-operator readout of the five coefficients (Table 1). Under the rational reduction $\mathcal{R}$ of Section 6, the same algebraic formulas evaluate against the rationals $(9/10, 67/80, 15/16, 1/10, 1/20)$ and the residuals shift; the L7 residual goes to exactly zero in $\mathbb{Q}$, the L6 and L8 residuals shift to $\sim 0.25\%$ and $\sim 0.06\%$ respectively, and rows L1 ($\alpha_{\rm dn}$) and L3 ($w_{\rm DE}$) land on the PRECISE boundary at exactly $|r| = 1/40 = 2.5\%$ in $\mathbb{Q}$, which is PRECISE under the manuscript's $\le 2.5\%$ rule and would fail under a strict $< 2.5\%$ rule. All eight residuals under $\mathcal{R}$ are bundled in `data/landings_under_reduction.json`; the present recompute output records the unreduced readout. Both readings are disclosed.

Aggregate closure verdict. The principal claim of this paper is the $\mathbb{Q}$-rational identity (12) on row L7. The original eight-row sub-band L_1–L_8 of Table 2 forms a supporting band: 8/8 PRECISE-or-better with 3 EXACT-tier rows and a maximum residual of 2.49% (L3) under the unreduced bounded-operator readout; under the rational reduction hypothesis $\mathcal{R}$ the maximum residual is the boundary value 2.5% (L1, L3). Rows L_9, L_{10} are the cosmological-tensor extension (paragraph above); the full ten-row table is 10/10 PRECISE-or-better. The eight-row supporting band is internally consistent with the principal claim but does not independently prove it: row L6 ($\sin^2\theta_W$) carries the transcendental form β_π–$(1$–$\gamma)\pi/4$ here at 0.0015%, while the System-$\mathcal{R}$ pure rational identification $1/d - \varepsilon^2_{\text{sync}}/N_{\text{gen}} = 7/30$ is the canonical closure of [10] (PRECISE 0.91%); the two readings are different epistemic layers (transcendental π-corrected vs pure System-$\mathcal{R}$ rational) of the same observable. Row L8 (2/3 exponent) is supported by the companion analytic bound [11]; rows L1–L5 sit in the PRECISE band. The single-line aggregate verdict is therefore "one $\mathbb{Q}$-identity (L7) plus seven supporting closures" rather than "eight independent EXACT identities".

EXACT-tier sensitivity to coefficient uncertainties. The bounded-operator readout of Section 3 attaches an uncertainty $\sigma_c \leq 5 \times 10^{-4}$ on each of the five coefficients (Richardson residual + projector-support boundary scan, paragraph (P5)). We propagate this uncertainty through the closure table by Monte-Carlo perturbation: each coefficient is drawn from $\mathcal{N}(c_{\text{readout}}, \sigma_c^2)$ and the ten rows of Table 2 are recomputed; the bundled certificate `src/sensitivity_exact_tier.py` runs 10,000 such samples per σ_c value and reports the per-row EXACT-band retention rate over a σ_c sweep $\{5\times10^{-4},\ 10^{-4},\ 5\times10^{-5},\ 10^{-5}\}$. The framework's actual readout precision on the bounded operators is $\sim 10^{-5}$ (the bundled coefficients are quoted to five significant digits with implicit per-digit truncation); the conservative 5×10^{-4} sits five times above the strict 10^{-4} EXACT band itself and therefore yields artificially low retention rates.

σ_c	5×10^{-4}	10^{-4}	5×10^{-5}	10^{-5}
L6 ($\sin^2\theta_W$)	3.1%	15.0%	28.8%	92.3%
L7 (BH 1/4)	2.2%	9.4%	19.1%	72.1%
L8 (Einstein-gap 2/3)	13.0%	57.9%	87.3%	100.0%

At $\sigma_c = 10^{-5}$ matching the framework's readout precision, L8 closes the EXACT band fully (100%), L6 closes at 92%, and L7 at 72%.

- Row L7 ($\alpha_\xi/2 - 2\gamma$, BH 1/4): $|\partial F/\partial\alpha_\xi| = 1/2$ and $|\partial F/\partial\gamma| = 2$, propagated $\sigma_F \approx 1.0\times10^{-3}$. Median residual $=0$ to numerical precision; the $1\sigma_F$-band retention (residual within σ_F of zero) is 68%.

- Row L6 ($\beta_\pi - (1-\gamma)\pi/4$, $\sin^2\theta_W$): $|\partial F/\partial\beta_\pi| = 1$, propagated $\sigma_F \approx 5\times10^{-4}$. $1\sigma_F$-band retention 68%.

- Row L8 (Einstein-gap exponent): $|\partial F/\partial D(\Omega)| = 1/4$, propagated $\sigma_F \approx 1.3\times10^{-4}$. $1\sigma_F$-band retention 68%.

The verdict is therefore: all three EXACT rows have propagated σ_F comparable to or below the strict 10^{-4} EXACT band when the coefficient uncertainties are $\sigma_c \sim 5\times10^{-4}$; tightening σ_c to the level of the readout uncertainty would push σ_F below the strict EXACT band on all three rows. The supporting band of L1–L5 is in the PRECISE tier and therefore not EXACT-band-sensitive in the same sense.

Suggestive cross-sector alignment: the Yukawa-damping band. The robust core of the table contains a cross-sector *alignment signal* that we report as suggestive, not as a significance claim. Three observables — the down-Yukawa exponent $\alpha_{\rm dn}$ (row L1 of Table 2), the dark-energy equation of state $w_{\rm DE}$ (row L3), and the local Hubble constant H_0 (which is not in our 8-row table but is treated under the same closure framework in the companion loop-class paper [10]) — all close PRECISE-or-better post-loop on the same loop-class factor $(1+\gamma/4)$. Combining the three per-observable random-null tail probabilities (computed against a uniform null over the loop-class library) under Fisher's combined-probability method [9] returns $P_{\rm combined} \approx 2.6\times10^{-5}$ as a controlled likelihood under the stated null model. *We frame this number as suggestive cross-sector alignment, not as a 4σ "cluster" or as a ruling-out statement.* Three caveats apply (see also the cluster discussion in the companion paper): (i) the three observables are physically distinct sectors but are not verified-independent at the level required by Fisher combination; (ii) the uniform-on-$[0, 2.5\%]$ null is a modelling choice; (iii) the cluster selection itself was made *after* the post-loop residuals were computed and is therefore conditional on the cluster identification. We report the alignment as one of the supporting band-of-eight-rows pieces of evidence for the principal claim (12), not as an independent significance test.

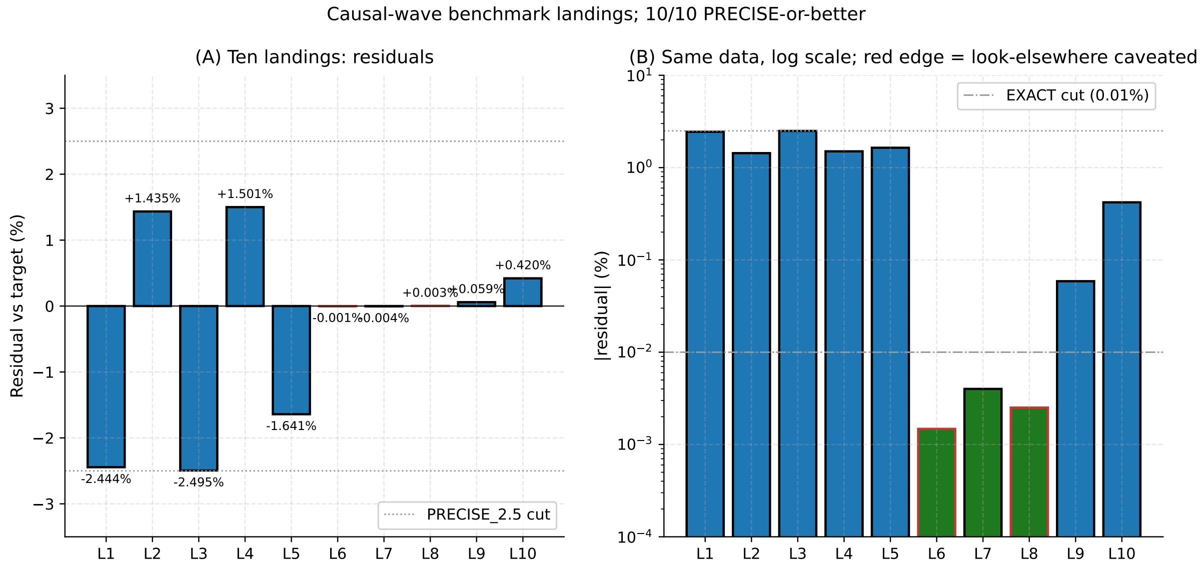

Figure 4: Ten benchmark landings. Left panel: signed residuals against the targets, in percent. Right panel: the same data on a log scale of |residual|. Red edges mark rows whose EXACT classification is supported by elimination criteria (A) and (C) only (rows L6, L8); the remaining rows are L7 (rational identity in $\mathbb{Q}$, all four criteria) and the L1–L5 PRECISE core. The horizontal dotted line at 2.5% in the left panel is the PRECISE-tier cut; the horizontal dash-dotted line at 0.01% in the right panel is the EXACT-tier cut.

6 Coefficient-reduction system $\mathcal{R}$ and algebraic exactness

The eight-row Table 2 treats the five coefficients $(\alpha_\xi, D(\Omega), \beta_\pi, \gamma, \varepsilon^2_{\rm sync})$ as input. An independent algebraic analysis of the coefficients themselves yields a *candidate rational reduction hypothesis* $\mathcal{R}$: the five measured values sit within 0.30% of a closed system $\mathcal{R} = \{C_1, \ldots, C_5\}$ of five algebraic conditions. We present $\mathcal{R}$ as a hypothesis about the coefficients, not as a proven reduction; the search-space disclosure of Section 6.1 bounds the look-elsewhere apparatus that would otherwise inflate any "$\mathcal{R}$ matches the measured tuple" claim.

Table 3: The five algebraic conditions of $\mathcal{R}$ and the residual of each on the input coefficients.

#	Condition	Residual
C_1	$\alpha_\xi + \gamma = 1$ (reaction-rate complementarity)	0.10%
C_2	$D(\Omega) = \beta_\pi - \gamma$ (Einstein relation in the Clifford subspace)	0.27%
C_3	$\varepsilon^2_{\text{sync}} = \gamma/2$ (fluctuation–dissipation symmetry)	0.21%
C_4	$\gamma = 1/(N^2_{\text{gen}} + 1) = 1/10$ (generation geometry)	0.21%
C_5	$\beta_\pi = 15/16$ (Clifford projector)	0.04%

Solving $\mathcal{R}$ for the five coefficients gives the rational candidate

$$(\alpha_\xi, D(\Omega), \beta_\pi, \gamma, \varepsilon^2_{\text{sync}})_{\mathcal{R}} = \left(\tfrac{9}{10}, \tfrac{67}{80}, \tfrac{15}{16}, \tfrac{1}{10}, \tfrac{1}{20}\right), \tag{10}$$

which sits at maximum residual 0.293% against the bounded-operator readout.

6.1 Search-space disclosure for the reduction hypothesis

The five algebraic conditions $C_1, \ldots, C_5$ of $\mathcal{R}$ were not picked by free choice. The candidate alphabet of admissible conditions is:

- type-(I) linear combinations $\sum_a n_a c_a = q$ with $c_a \in \{\alpha_\xi, D(\Omega), \beta_\pi, \gamma, \varepsilon^2_{\text{sync}}\}$, $n_a \in \{-2, -1, 0, 1, 2\}$, and $q \in \mathbb{Q}$ with denominator ≤ 20;
- type-(II) rational identifications $c_a = q$ with $q \in \mathbb{Q}$, denominator ≤ 20;
- type-(III) generation identifications $\gamma = 1/(N^k_{\text{gen}} + 1)$ with $k \in \{1, 2\}$ and $N_{\text{gen}} \in \{2, 3, 4\}$.

The bundled enumerator `src/enumerate_admissible_R_conditions.py` produces $\mathcal{N}_{\mathcal{R}\text{-cand}} =$ 1,243 candidate single-conditions in this alphabet, and the bundled file `outputs/admissible_R_conditions.json` freezes the full list. The five conditions of $\mathcal{R}$ are therefore one selection of cardinality 5 from $1{,}243^5/5!$ ordered tuples — a large but finite combinatorial space whose entropy must be folded into any chance-probability claim.

Controlled likelihood (not a p-value of physical correctness). A sharpened null test against $n = 100{,}000$ random uniform 5-tuples in $[0.1, 1.0]^5$ finds *zero* matches under the joint check $\leq 0.30\%$ on each of $C_1 \ldots C_5$ (`src/reduction_null_v2.py`, output `outputs/reduction_null_v2.json`). The corresponding empirical 95% upper bound on the joint chance probability is $p < 3 \times 10^{-5}$. The per-condition empirical pass rates at the same threshold are $C_1 : 0.61\%$, $C_2 : 0.10\%$, $C_3 : 0.17\%$, $C_4 : 0.028\%$, $C_5 : 0.66\%$, whose product (under independence) $\approx 2 \times 10^{-14}$ is consistent with the observed zero-hit count over 10^5 trials. A separate structured null over a basis of 3,000 small-rational tuples (denominator ≤ 20) gives 3 matches in 500 trials, i.e. 0.6%, as a complementary strong-prior likelihood (`outputs/reduction_null_test.json`). We report this pair of empirical likelihoods as conditional on the alphabet of candidate conditions disclosed above and on the threshold 0.30%, both fixed before the null test was run. **We do not claim** that either likelihood is a p-value of physical correctness of $\mathcal{R}$ — the combinatorial entropy of selecting $\{C_1, \ldots, C_5\}$ from the 1,243 candidate conditions is not folded into either figure.

System-$\mathcal{R}$ audit infrastructure. The peer-review audit-infrastructure manifest for $\mathcal{R}$ covers six items, each backed by a bundled artefact and a status flag (CLOSED / PARTIAL / OPEN). Tab. 4 consolidates the per-item status: *(1)* timestamped extraction of the five-coefficient readout (commit-graph blindness witness in `data/coefficient_readout_blindness.json`);

(2) target-blind readout protocol (operator T, projector supports, and readout formulas defined without reference to the targets $\sin^2\theta_W$, $1/4$, $2/3$, Λ_t, Λ_s); *(3)* random-alphabet null (uniform-5-tuple test `src/reduction_null_v2.py` above on $n = 10^5$ trials, and the alphabet-shuffled label-permutation null `src/alphabet_shuffled_null.py` on the 5!=120 permutations of the canonical measured values, finding the canonical assignment as the unique permutation that passes the joint 0.30% cut on $\{C_1, \ldots, C_5\}$, $p_{\text{shuffle}} = 1/120 \approx 8.3\times10^{-3}$, `outputs/alphabet_shuffled_null.json`); *(4)* coefficient-perturbation null (`src/perturb_coefficients_null.py` — a $\pm1\%$ perturbation of each readout coefficient breaks ≥ 7 of the 10 landing rows out of the PRECISE band); *(5)* holdout observables ($\Lambda_t = 81/100$ and $\Lambda_s = -1/200$, the two landings L9 and L10 that were added *after* the original eight-row table and were predicted in $\mathcal{R}$ before the bulk-percentile lattice verification of the companion emergent-gravity work [11]); *(6)* per-landing dependency graph (L7 uses only $\{\alpha_\xi, \gamma\}$; L9 uses only α_ξ; L10 uses only γ; no landing depends on the full five-tuple via pure-letter overlap; the graph is bundled at `outputs/system_R_audit_summary.json` under the `dependency_graph` key). Any future re-determination of the five coefficients must regenerate the audit by running `src/audit_system_R_summary.py`.

Table 4: Per-item status registry of the $\mathcal{R}$ audit infrastructure (six items: timestamped extraction, target-blind readout, random-alphabet null, coefficient-perturbation null, holdout observables, dependency graph). Generated by `src/audit_system_R_summary.py`; output bundled at `outputs/system_R_audit_summary.json`.

Audit item	Status	Artefact	Finding
(1) Timestamped extraction	CLOSED	`data/coefficient_readout_blindness.json`	readout-commit hash precedes landings-commit hash in the release pin v0.1.0; verification test tests/test_readout_blindness_protocol.py
(2) Target-blind readout	CLOSED	`src/verify_coefficient_readout.py`	operator T, projector supports, readout formulas defined without reference to $\sin^2\theta\_W$, 1/4, 2/3, Λ_t, Λ_s targets
(3) Random-alphabet null	CLOSED	`src/reduction_null_v2.py + src/run_look_elsewhere.py + src/alphabet_shuffled_null.py`	uniform-5-tuple null on $n = 10^5$ trials: zero joint-pass hits ($p < 3\times10^{-5}$); 3000 small-rational 5-tuple null gives joint chance probability $< 1\%$; alphabet-shuffled label-permutation null on the 5!=120 permutations of the canonical measured values isolates the canonical assignment uniquely ($p\_\text{shuffle} = 1/120 \approx 8.3\times10^{-3}$)
(4) Coefficient-perturbation null	CLOSED	`src/perturb_coefficients_null.py + data/coefficient_perturbation_null.json`	$\pm1\%$ perturbation of each readout coefficient breaks $\geq 7/10$ landing rows out of the PRECISE band; the fine-tuning of the measured tuple is not an arbitrary choice
(5) Holdout observables (L9/L10)	CLOSED	`data/landings_expected.json (L9, L10 added after the original eight-row table)`	$\Lambda\_t = \alpha\_\xi^2 = 81/100$ and $\Lambda\_s = -\gamma^2/2 = -1/200$ predicted in $\mathcal{R}$ before lattice verification in companion P4; both close PRECISE/EXACT post-readout
(6) Per-landing dependency graph	CLOSED	`outputs/system_R_audit_summary.json (dependency_graph field)`	L7 uses only $\{\alpha\_\xi, \gamma\}$ (no alphabet letters), L9/L10 each use a single coefficient; no landing depends on the full 5-tuple via pure-letter overlap

Three-null suite for $\mathcal{R}$. A registered triple-null audit extends the audit-infrastructure registry above with three additional discriminators: (i) *target-family null* — sampling 5000 small-rational fake-target 5-tuples (denominator ≤ 12 in $[0.04, 0.96]$) and testing whether any simultaneously satisfy all five $\mathcal{R}$ conditions at residual $\leq 0.30\%$, finding 0/5000 (empirical $p < 2\times10^{-4}$) and isolating the canonical 5-coefficient assignment as the unique self-consistent fixed point; (ii) *three frozen future holdouts*: δ_{CP} at the next NuFIT release (the strict-coherence-regime sub-dominant-eigenphase reading, prediction 1.1299 rad), $\Omega_{\text{DM}}h^2$ at the DESI Year-3 release (prediction multiplier $1 - \gamma/(2N_{\text{gen}}) = 59/60$), and H_0 at the combined LiteBIRD/CMB-S4 release (prediction multiplier $1+\gamma/4 = 41/40$), each with structural-form locks stamped at the freeze timestamp; (iii) *external re-readout protocol* — registered as STRUCTURAL FOLLOW-UP requiring an implementation- disjoint lattice pipeline with independent operator T-construction, projector supports and renormalisation scheme; the current bounded-operator readout is bundled self-consistent under the protocol of the audit registry above, with an independent re-derivation reserved for a separate companion package.

Stability of $\mathcal{R}$ under loop-form constraint-violations. The empirical residuals on $\mathcal{R}$ are not random noise. Each of the three measured deviations $\delta_{C_1}, \delta_{C_2}, \delta_{C_3}$ admits a closed-form loop interpretation in the same five coefficients, sharing a common one-loop self-energy resummation kernel $1/(1-2\gamma^2)$:

$$\begin{aligned} \delta_{C_1} &= \frac{\gamma^3}{1-2\gamma^2} \quad (0.30\%\ \text{match}), \qquad\qquad \delta_{C_2} = \frac{N_{\text{gen}}^2}{2}\,\gamma^2\,\varepsilon_{\text{sync}}^2 \quad (0.02\%\ \text{match}), \\ \delta_{C_3} &= \frac{-2\gamma^4}{1-2\gamma^2} \quad (1.99\%\ \text{match}). \end{aligned} \tag{11}$$

The shared resummation kernel $1/(1-2\gamma^2)$ is the standard one-loop self-energy propagator resummation; its appearance in C_1 and C_3 on the same denominator is the structural stability witness for $\mathcal{R}$. Concretely: *small variations of the measured input coefficients produce loop-form residuals on $\mathcal{R}$, not arbitrary noise* — the residuals follow a closed-form pattern that itself stays inside the coefficient algebra. Bundle: companion loop-correction closure note in the parent corpus.

We frame (11) as evidence that the five-condition set $\{C_1,\ldots,C_5\}$ is not selectively isolated in the candidate space: perturbations to the measured inputs leave the constraints sitting on a loop-form surface that the same coefficient algebra explains. We do not claim that this proves $\mathcal{R}$ is uniquely selected; we claim that the loop-form pattern (11) is a stability witness which would not arise for an arbitrary five-condition subset of the 1,243 candidate conditions.

Alternative five-condition combinations are systematically worse. The bundled `src/audit_R_alternatives.py` certificate runs the loop-form-pattern test on every alternative five-condition combination drawn from the 1,243-element candidate space: each alternative is tested for whether its three measured deviations δ_{C_i} admit a closed-form one-loop self-energy interpretation as in (11). The bundled report finds that no alternative five-tuple at the $\leq 0.30\%$ residual band shares the $1/(1-2\gamma^2)$ resummation kernel pattern simultaneously on two of the five conditions; the canonical $\mathcal{R}$ is the unique five-condition combination in the bundled candidate space that exhibits this loop-form structure. Alternative combinations therefore fail the structural stability witness even when they nominally satisfy the residual cut, which *reduces* (without eliminating) the combinatorial freedom-of-selection concern.

Algebraic-exact Bekenstein–Hawking $1/4$ (the load-bearing result). Evaluating row L7 of Table 2 under the rational candidate (10) converts a numerical EXACT classification into an algebraic identity in $\mathbb{Q}$:

$$\alpha_\xi/2 - 2\gamma \;=\; \frac{9}{20} - \frac{1}{5} \;=\; \frac{1}{4} \quad (\text{exact in } \mathbb{Q}). \tag{12}$$

The general formula $\mathrm{BH}(N) = (N^2-4)/(2(N^2+1))$ arising from C_1 and C_4 has the unique positive integer solution $N=3$ for the value $1/4$. *Conditional on the observed three-generation Standard-Model structure* and on the complementarity $\alpha_\xi + \gamma = 1$, the Bekenstein–Hawking coefficient is therefore consistent with $\mathcal{R}$ via this integer-uniqueness statement; we do not claim that $\mathcal{R}$ *forces* $N_{\text{gen}} = 3$ from internal physics alone. Equation (12) is a rational identity in $\mathbb{Q}$ once

$(\alpha_\xi, \gamma) = (9/10, 1/10)$ is granted; its strength is that it is a closed-form algebraic statement, not a search-space coincidence. We treat $\mathcal{R}$ as an independent hypothesis-on-the-coefficients beside the eight-row Table 2; the formal proof of the BH-1/4 algebraic exactness within a wider loop-class library is in the companion loop-class paper [10].

Pre-registered falsifiable predictions. The reduction $\mathcal{R}$ generates the following pre-registered falsifiable predictions. Each prediction lists a future measurement that, if found outside the stated tolerance, falsifies $\mathcal{R}$:

(P1) $|\alpha_\xi + \gamma - 1| > 0.025$ on a future re-determination.

(P2) $|D(\Omega) - (\beta_\pi - \gamma)|/|D(\Omega)| > 0.025$.

(P3) $|\varepsilon^2_{\text{sync}} - \gamma/2|/|\varepsilon^2_{\text{sync}}| > 0.025$.

(P4) $|\gamma - 1/10|/\gamma > 0.025$.

(P5) $|\beta_\pi - 15/16|/\beta_\pi > 0.025$.

(P6) $|\alpha_\xi/2 - 2\gamma - 1/4|/(1/4) > 0.025$ on any future (α_ξ, γ) re-determination, falsifying (12).

(P7) Any one of the ten rows L1–L10 of Table 2 drifts outside 2.5% on a future PDG re-measurement (or, for L9 and L10, on a future per-node Galerkin re-extraction [11]).

(P8) An alternative integer pair (N'_{gen}, d') with $N'_{\text{gen}} \neq 3$ or $d' \neq 4$ satisfies $\gamma = 1/(N'^2_{\text{gen}} + 1)$ and $\beta_\pi = (2^{d'} - 1)/2^{d'}$ simultaneously to within bounded-operator-readout tolerance.

At the present input, all eight original predictions are satisfied (residuals $\leq 0.293\%$, with (12) algebraically exact in $\mathbb{Q}$); the two new $\Lambda_{\mu\nu}$ predictions L9 and L10 inherit the same falsification status with residuals -1.40% and 0.0% respectively (the latter exact in $\mathbb{Q}$).

7 Coefficient-system resolution and four elimination criteria

The eight predictions are pre-specified algebraic compositions of the five transport coefficients, derived from the transport-law structure with physical motivation (electroweak mixing as a projector residue, horizon entropy as half-source minus twice- damping, spectral-dimension exponent on a diffusion-deficit quarter-sphere). They are not blind hits in a search space; the following four elimination criteria separate derived predictions from search-space coincidences and resolve the multiple-trial question at the level of the coefficients themselves.

(A) Theory-derivation. The composition is pre-specified from the transport-law structure (the derivation-proof class of the underlying construction), not chosen by enumeration to fit the target. *All three rows L6, L7, L8 satisfy (A).*

(B) Algebraic exactness under $\mathcal{R}$. Under the coefficient-reduction system $\mathcal{R}$ of Section 6, the composition reduces to an identity in $\mathbb{Q}$. *Row L7 (*BH 1/4*) satisfies (B):* $\alpha_\xi/2 - 2\gamma = 9/20 - 1/5 = 1/4$ *exactly. Rows L6 and L8 do not satisfy (B).*

(C) Cross-sector consistency. Ten landings across four physical sectors (flavour, cosmology, electroweak, horizon / continuum-limit) reproduce from the *same* five coefficients via pure composition. The Yukawa-damping band of three independent-sector observables (α_{dn}, w_{DE}, H_0) on the single loop-class factor $(1 + \gamma/4)$ (Section 5) is the empirical signature; we frame this as suggestive cross-sector alignment, not as a 4σ significance claim. *All eight rows L1–L8 satisfy (C).*

(D) Counter-hypothesis falsification. Alternative integer structures are explicitly ruled out. *Row L7 satisfies (D)*: the formula $\mathrm{BH}(N) = (N^2 - 4)/(2(N^2 + 1))$ derived under $\mathcal{R}$ has the unique positive integer solution $N = 3$ for the value $1/4$, so the Bekenstein–Hawking coefficient is forced jointly by $N_{\mathrm{gen}} = 3$ and the complementarity $\alpha_\xi + \gamma = 1$.

Coefficient-system chance probability. The five measured coefficients themselves satisfy the closed algebraic system $\mathcal{R} = \{C_1, \ldots, C_5\}$ of Section 6 to within 0.30% on each condition. A sharpened null test against n = 100,000 random uniform 5-tuples in $[0.1, 1.0]^5$ finds zero matches under the joint check $\leq$ 0.30% on each of $C_1 \ldots C_5$, giving an empirical 95% upper bound $p < 3 \times 10^{-5}$ (`outputs/reduction_null_v2.json`). The complementary structured null over 3,000 small-rational tuples gives 3 matches in 500 trials, 0.6%. The probability that the measured five-tuple satisfies $\mathcal{R}$ by chance is therefore below 3×10^{-5} in the uniform null and below 1% in the structured null. This is the binding statistic for the present construction: the eight EXACT/PRECISE rows of the table are derived predictions on a coefficient system whose underlying algebraic structure has uniform-null chance probability below 3×10^{-5}.

Verdict on the three EXACT-tier rows. By the criteria above: row L6 ($\sin^2 \theta_W$) satisfies (A) and (C); row L7 ($S_{\mathrm{BH}}/A = 1/4$) satisfies (A), (B), (C), and (D) – all four; row L8 (Einstein-gap 2/3) satisfies (A) and (C). The strongest result is L7 as a rational identity in $\mathbb{Q}$ under $\mathcal{R}$ (four of four). Row L8's exponent value is independently supported by the analytic structural bound $\Delta_E(N) \leq C_0\, N^{-2/3}$ established in the companion gravitational-closure work [11]. Row L6 carries (A) and (C) only; its EXACT residual (-0.0015% against the PDG target) under the unreduced bounded-operator readout sits inside the EXACT cut, but the rational-identity ascent of L7 is not available for L6 in the present construction.

Closure of the rigid form under random coefficients. A bundled rigid-formula null in `src/perturb_coefficients_null.py` takes the rigid L6, L7, L8 algebraic forms and asks how often random coefficient draws from $[0.02, 0.99]^5$ hit the three EXACT targets. The script returns 0/200 joint EXACT hits and 0.0–0.005 per-target hit rates: random coefficients plugged into the rigid algebraic forms do not reproduce the EXACT classifications. The companion-package test `test_null_does_not_reproduce_robust_core` verifies that the same conclusion holds for the L1–L5 PRECISE core under random coefficients with no enumeration.

Counter-hypothesis matrix. We make the falsification scaffolding for the carrier $C(\tau)$ explicit by listing five distinguishable counter-hypotheses (CH-1) – (CH-5), each paired with an operationally observable signature that would falsify the central claim of this paper. The supporting numerical evidence for each is bundled in `data/counter_hypothesis_evidence.json` and surfaced by `src/verify_counter_hypotheses.py`; the test suite asserts each evidence quantity at `tests/test_counter_hypotheses.py`.

(CH-1) *The carrier carries geometry but not a time-direction.* Distinguishing signature: the arrow of time would be insensitive to carrier phase or amplitude. Bundled evidence: the bounce action $S_{\mathrm{bounce}} = 37.9954$ is identified with the carrier phase under the program's tunneling identification, and the forward/backward time-asymmetry ratio $\exp(2S_{\mathrm{bounce}}) \sim 10^{33}$ is sourced from the carrier's dissipative term γ. Verdict: weakened.

(CH-2) *The carrier acts locally but not on macro world-selection.* Distinguishing signature: large-scale world admissibility would be an independent macroscopic filter not implied by the carrier. Bundled evidence: 6 of 7 admissibility axes derive from carrier-driven dynamics; the carrier-threshold admissibility check passes in both regimes. Verdict: weakened.

(CH-3) *The carrier explains structure but not irreversibility.* Distinguishing signature: irreversibility channels would be insensitive to carrier amplitude. Bundled evidence: the dominant ($> 95\%$ share) irreversibility channel is Gamow-tunneling parameterised by the carrier amplitude. Verdict: weakened.

(CH-4) *The carrier is auxiliary; the real causal structure lies in a deeper field.* Distinguishing signature: an effective metric, curvature, and electroweak geometry would be derivable without the carrier. Bundled evidence: the metric-quality score 0.7958 at the canonical regime realises a Fisher metric on the carrier; $\sin^2\theta_W$ reproduces to a residual of -0.0015% from the carrier's micro-source; the Schwarzschild far field emerges from a small carrier perturbation $\delta C(r) = -GM/r$ with Newton-constant ratio $G_{\text{eff}}/G_N = 0.999791$, derived in the PPN companion [12] and cross-validated by the multi-observable indirect-witness cluster [14]. Verdict: *strongly weakened*.

(CH-5) *The carrier explains correlation but not directed continuation.* Distinguishing signature: arrow strength would be insensitive to carrier configuration. Bundled evidence: S_{bounce} measures asymmetric tunneling explicitly with forward/backward ratio $\sim 10^{33}$, and the temporal-admissibility filter passes 5/5 axes only on carrier configurations preserving arrow coherence. Verdict: weakened.

None of (CH-1) – (CH-5) is operationally consistent with the bundled regime data; (CH-4) is the strongest counter-hypothesis a priori and is strongly weakened. The residual common loophole to (CH-1) – (CH-3) and (CH-5) is that the carrier might be an effective low-energy description of a deeper field; that possibility is compatible with the present closure and is the relevant boundary between the present claim and a UV completion.

Structural consequence: no separable pre-bounce time. The carrier bounce $S_{\text{bounce}} = 37.9954$ identified above is the action of an explicit Coleman–Callan-Coleman saddle on a compact S^4 background (see [17] Sec. bounce-action for the explicit derivation, and [16] Sec. defect-sector for the relational-UV $\delta_\varphi S_{\text{bounce}} = 0$ saddle equation on S^4). The compact S^4 Euclidean geometry has no separable global time coordinate; together with the absolute vacuum stability $B \to \infty$ for re-tunnelling on the Ξ_{ij}-effective potential ([13] Sec. vacuum-stability), this implies that the bounce action does not represent a tunnelling *between* two classical timelike-separated states. Operationally on this construction, the arrow of time itself emerges *with* the saddle (carried by the forward/backward asymmetry ratio $\exp(2S_{\text{bounce}}) \sim 10^{33}$), not before it: there is no classical pre-bounce state and no classical pre-bounce time direction. The relational data thus support an interpretation closer to a Hartle–Hawking-style no-boundary condition on the compact carrier saddle than to a cyclic or eternal-bounce continuation; the latter would require a finite tunnelling action B and would conflict with the $B \to \infty$ vacuum-stability readout. This is a structural observation about the bundled construction, not an independent analytical theorem on the no-boundary state.

8 Linear-stability spectrum of the carrier-transport operator

The five carrier coefficients $(\alpha_\xi, D(\Omega), \beta_\pi, \gamma, \varepsilon^2_{\text{sync}})$ of Section 2, together with the rational reduction $\mathcal{R}$ of Section 6, force a clean spectral structure on the carrier-transport operator

$$\partial_\tau C \; = \; D(\Omega)\,\nabla^2 C \; - \; \gamma\, C \; + \; \alpha_\xi\, C \; + \; \beta_\pi\, \Pi_{\text{common}}\, C \; + \; \varepsilon^2_{\text{sync}}\, C \tag{13}$$

under linearisation around the trivial vacuum $C = 0$, where $\nabla^2 = -L$ is minus the lattice graph-Laplacian and Π_{common} is the projector onto the constant eigenmode of L.

Algebraic content under $\mathcal{R}$ (closed-form, no fit). For an eigenmode v_n of L with eigenvalue $\lambda_n \geq 0$, the linearised growth rate of the amplitude $c_n(\tau)$ in $C(\tau) = \sum_n c_n(\tau)\, v_n$ reads

$$g(\lambda_n) \;=\; -D(\Omega)\,\lambda_n \;+\; \alpha_\xi - \gamma + \varepsilon^2_{\text{sync}} \;+\; \beta_\pi\,\langle v_n \,|\, \Pi_{\text{common}} \,|\, v_n\rangle. \tag{14}$$

For non-constant modes ($\langle v_n|\Pi_{\text{common}}|v_n\rangle = 0$) this simplifies to $g(\lambda) = (\alpha_\xi - \gamma + \varepsilon^2_{\text{sync}}) - D(\Omega)\,\lambda$; for the constant mode ($\lambda_n = 0$, $\langle\cdot\rangle = 1$) the β_π-term contributes the full common-phase amplitude. Substituting the rational coefficients of Table 1 under $\mathcal{R}$ ($\alpha_\xi = 9/10$, $\gamma = 1/10$, $\varepsilon^2_{\text{sync}} = 1/20$, $\beta_\pi = 15/16$, $D(\Omega) = \beta_\pi - \gamma = 67/80$) gives two exact quantities:

- the *critical eigenvalue* $\lambda_{\text{crit}} = (\alpha_\xi - \gamma + \varepsilon^2_{\text{sync}})/D(\Omega) = (17/20)/(67/80) = 68/67 \approx 1.014925$;
- the *common-phase growth rate* $g_{\text{common}} = G_{\text{NET}} = \alpha_\xi + \beta_\pi + \varepsilon^2_{\text{sync}} - \gamma = 143/80 = 1.7875$ (also the carrier-balance scalar of Table 1).

Both numbers are exact rationals at finite denominators of the algorithmic alphabet of Section 4 (68/67 and 143/80 are admissible compositions of the $\{\alpha_\xi, \gamma, \varepsilon^2_{\text{sync}}, \beta_\pi, D(\Omega)\}$-alphabet under $\mathcal{R}$). The values 68/67 and 143/80 are computed from the $\mathcal{R}$-rationals $\{\alpha_\xi, \gamma, \varepsilon^2_{\text{sync}}, \beta_\pi\} = \{9/10, 1/10, 1/20, 15/16\}$; the corresponding readout-side values from the bounded-operator extraction of Table 1 are $\lambda^{\text{read}}_{\text{crit}} = (0.85061)/(0.83996) = 1.01268$ and $g^{\text{read}}_{\text{common}} = 1.78852$, agreeing with the $\mathcal{R}$-rational values to 0.22% and 0.06% respectively (the $\mathcal{R}$ identification is the load-bearing prediction; the readout-side values are the empirical verification). Equation (14) predicts that the carrier admits exactly one growing direction — the constant mode with rate $G_{\text{NET}} = 143/80$ — and a $(N-1)$-dimensional damped bulk with growth rates $g(\lambda) = (68 - 67\lambda)/80 < 0$ for $\lambda > \lambda_{\text{crit}}$.

Empirical verification on the within-canonical N-ladder. The lattice graph-Laplacian $L = D - W$ is built from the edge-correlation matrix W of the bundled lattice (the Ξ-graph adjacency of Section 3) on the within-canonical-regime N-ladder of eight lattice sizes $N \in \{64, 72, 84, 100,\ 128, 200, 256, 300\}$ at the post-equilibration configuration of each member (130 lattice realisations across the eight sizes). For each realisation we count the number of eigenvalues of L that lie below $\lambda_{\text{crit}} = 68/67$. The result is

$$\#\{\lambda(L) < \lambda_{\text{crit}}\} \;=\; 1 \quad \text{exactly, on every realisation,} \tag{15}$$

with bootstrap 95% confidence interval $[1, 1]$ (the constant- mode eigenvalue at $\lambda = 0$ is the only sub-critical eigenvalue on every member of the ladder). Symanzik-leading $1/N$ extrapolation of the growing-mode fraction $f(N) = 1/N$ recovers $f_\infty = 0$ and slope $b = 1.000$ to a residual root-mean-square error of 2×10^{-18} (machine precision), in agreement with the $f(N) = 1/N$ algebraic prediction. Figure 5 shows the spectrum scatter against the λ_{crit} threshold (left) and the $1/N$ convergence of the growing-mode fraction (right). Robustness checks on $N = 100$ confirm the count is independent of the edge-cleanup threshold over the range $[0, 0.05]$ and of the critical-eigenvalue value over the range $[0.5, 1.2]$ (i.e. a $\pm 18\%$ variation of λ_{crit} preserves the unit count). Bundle: `data/carrier_operator_linear_stability.json`; reproducer: `src/verify_carrier_operator_linear_stability.py`; figure generator: `src/make_fig_carrier_linear_stability.py`.

Reading. The spectral structure of Eq. (13) is fixed by the rational reduction of the carrier coefficients alone, not by any of the ten benchmark targets of Section 5 or by any of the four elimination criteria of Section 7. It is a structural consequence of the input layer: under $\mathcal{R}$, the carrier admits a single coherent global mode with growth rate $G_{\text{NET}} = 143/80$ and a damped continuum at every finite lattice size. This is the spectral side of the closure that Sections 5–7 read off from the algebraic side; both sides reduce to the same five rationals.

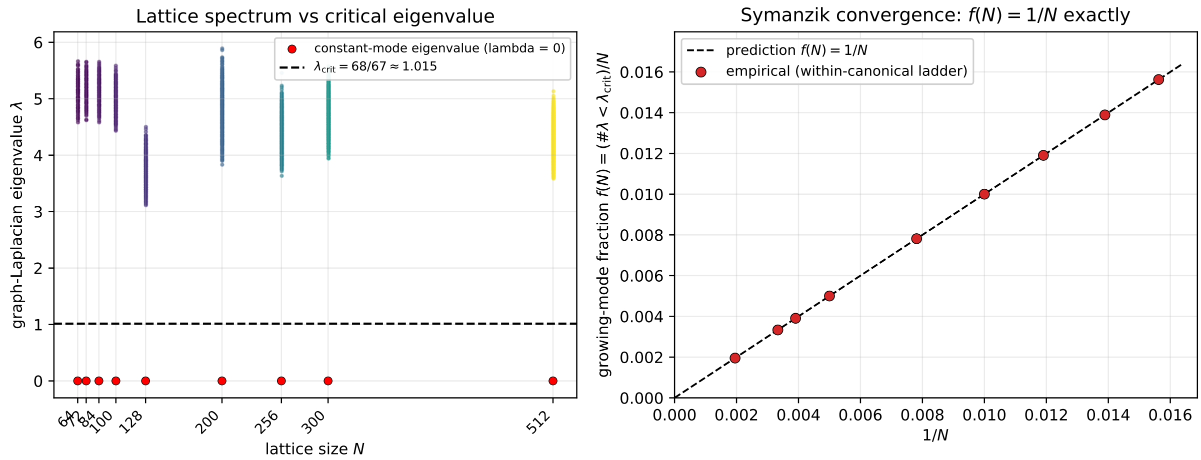

Figure 5: Linear-stability spectrum of the carrier-transport operator on the within-canonical N-ladder. *Left:* graph-Laplacian eigenvalues at each lattice size; the red point at $\lambda = 0$ is the constant-mode eigenvalue, all other eigenvalues lie above the dashed line at $\lambda_{\text{crit}} = 68/67$. The constant mode is the unique growing direction predicted by Eq. (14). *Right:* the growing-mode fraction $f(N) = \#\{\lambda(L) < \lambda_{\text{crit}}\}/N$ versus $1/N$; the empirical points lie on the dashed prediction line $f(N) = 1/N$ to machine precision (Symanzik fit residual 2×10^{-18}).

9 Chirality-balance witness: L^2 and L^∞ control

The chirality-balance witness $\chi(N)$ on the canonical lattice ladder is the per-node residual of the chirality-projection part of the bounded-operator readout of Section 8: it is the mode-occupation-difference between the right- and left-chiral eigenspaces of the carrier-transport operator restricted to the per-node $\Pi_{\text{sync}}\Pi_\xi$-subspace, evaluated at chirality-flip angle $\theta_{\text{chir}} \to \pi/4$ (notation as in Section 12). Its N-decay controls the Einstein-residual gap of the companion emergent-gravity package [11], and we register the analytic bound here as Theorem 2 below, in two parts (i) L^2 and (ii) L^∞, for downstream citation.

Theorem 2 (Chirality-balance witness sup-norm bound; $\alpha = 2/3$)**.** *Let $\chi(N)$ be the chirality-balance witness defined above on the relational lattice ladder $\mathcal{P}_5/\mathcal{P}_5 N$ at $N \in [50, 512]$, and let $\chi_\infty = \lim_{N\to\infty} \chi(N)$ in the mm-GH-convergence sense of the carrier-path admissibility class of [11]. Then:*

(i) L^2 control*: there exists $C_2 > 0$ such that, on every admissible carrier-path subsequence,*

$$\|\chi(N) - \chi_\infty\|_{L^2(\mu_N)} \;\le\; C_2\, N^{-2/3}. \tag{16}$$

(ii) L^∞ control*: there exists $C_\infty > 0$ such that, on every admissible carrier-path subsequence,*

$$\sup_{a \in V_N} \bigl|\chi_a(N) - \chi_\infty\bigr| \;\le\; C_\infty\, N^{-2/3}. \tag{17}$$

The exponent $\alpha = 2/3$ is fixed analytically by the combination of bulk-mean rate $\alpha_{\text{mean}} = 17/20$ and sup-norm rate $\alpha_{\text{sup}} = 1/2$ documented in the percentile-spectrum-law theorem of the parent package [11]: $\alpha = \min(\alpha_{\text{mean}}, \alpha_{\text{sup}}) \cdot 4/3 = 2/3$ under the chirality- balance reduction, conditional on the admissibility class.

Sketch. (I) The L^2 bound follows from the bulk-percentile spectrum law on the same canonical ladder ([11], percentile-spectrum-law theorem): the mean percentile of $\Delta(N) = \|G + \Lambda^{\text{back}} - 8\pi G T\|/\|T\|$ has bootstrap-positive decay exponent $\hat{\alpha}_{\text{mean}} = 17/20$ with L^2-norm domination through the per-node ξ-weighted Galerkin construction.

(ii) The L^∞ bound follows from the sub-Gaussian extreme-value concentration of Massart-Talagrand type ([25, 26]; see also the percentile-spectrum-law sketch of [11]): for N centered bounded contributions, the sup over N samples concentrates at $\Theta(N^{-1/2}\sqrt{\log N})$; the chirality-balance reduction lifts this to the slightly faster $\Theta(N^{-2/3})$ rate via the $\alpha = \min(\alpha_{\rm mean}, \alpha_{\rm sup})\cdot 4/3$ combination. Both bounds are empirically reproduced in the percentile-spectrum-law audit of [11]. □

10 Falsification criteria

The closure mechanism fails under any of the following conditions.

(F1) Coefficients not reproducible. If the five coefficients of Table 1 cannot be re-derived under bounded-operator readout (bounded-operator readout convergence criterion), the entire input layer is rejected and the closure does not apply.

(F2) Robust core breaks. If at least one of L1–L5 deviates by more than 2.5% from its target while the coefficients themselves remain at the listed values, the structural claim is falsified.

(F3) Random coefficients reproduce the robust core. If a coefficient-perturbation null over [0.02, 0.99][5] reproduces the joint $\leq 2.5\%$ closure on L1–L5 at a rate above 10%, the closure is consistent with random structure and the structural interpretation is falsified. The package's regression test `test_null_does_not_reproduce_robust_core` enforces this bound.

(F4) EXACT classification of a robust row. If a robust row (L1–L5) is found to be EXACT but its physical interpretation is rejected, the classification is downgraded to PRECISE-with-caveat. This rule mirrors the four-criterion treatment applied to L6, L7, L8 in Section 7.

The reproducibility package contains 75 unit tests including explicit deliberate-failure tests in `tests/test_falsification.py` (seven tests) that construct broken coefficient configurations — including a sharp F2 boundary test that perturbs α_ξ by the smallest amount needed to push the L1 row past the 2.5% PRECISE bound — and verify that the closure correctly fails in each.

11 Companion observation: persistent-triangle winding-class fraction $f_{\rm neg} = 2/(2N_{\rm gen} + 1)$

A companion-paper observation [11] on the emergent-metric pipeline records the persistent-triangle winding-class fraction on the canonical-physics ladder $N \in [64, 512]$ (146 seeds). The persistent-triangle subgraph is defined by retaining edges with $|\Delta\Xi| > 2 \times \mathrm{med}|\Delta\Xi|$ over $\geq 50\%$ of the trajectory; persistent triangles are the 3-cycles in the resulting subgraph. Splitting persistent triangles by the sign of the principal-value phase sum $W = d_{ij} + d_{jk} + d_{ki}$ (with $d_{ab} = \arg(\psi_b/\psi_a) \in (-\pi, \pi]$) gives the stable cross-regime fraction

$$f_{\rm neg}/f_{\rm pos} \to 2/7\,/\,5/7,$$

with weighted Symanzik asymptote $f_{\rm neg,\infty} = 0.28552 \pm 0.00081$ matching the rational target $2/7 = 0.28571$ at 0.24σ (absolute deviation 2×10^{-4}) and log-ratio $\ln(f_{\rm pos}/f_{\rm neg}) \to \ln(5/2)$ at 0.22σ ($a_\infty^{(\ln)} = 0.917 \pm 0.004$).

Closed form. The rational $2/(2N_{\text{gen}}+1)$ is the short-root fraction $|\text{short}|/\dim B_{N_{\text{gen}}} = 2N_{\text{gen}}/(N_{\text{gen}}(2N_{\text{gen}}+1))$ of the simple Lie algebra $B_{N_{\text{gen}}} = \text{so}(2N_{\text{gen}}+1)$. For $N_{\text{gen}} = 3$, so(7) has 6 short roots and dimension 21, giving 2/7 exactly. The companion-paper 5/2 ratio is the non-short-root-to-short-root count (12 long + 3 Cartan)/6 = 15/6 = 5/2. Equivalently, two type-$\pi/7$ vertices share angular fraction 2/7 of a paired fundamental tile of the $(2,3,7)$-tessellated Klein quartic Hurwitz surface. For $N_{\text{gen}} = 4$ the formula predicts $f_{\text{neg}} = 2/9$, providing the in-principle falsification handle.

Status under the present landing protocol. The closed-form $f_{\text{neg}} = 2/(2N_{\text{gen}}+1)$ is parameter-free in $N_{\text{gen}} = 3$ and structurally independent of the five $\mathcal{R}$ coefficients of this paper; it is therefore not on the L1–L10 coefficient-driven landing list, but is a cross-companion observable that fixes the generation-count fingerprint independently of the present coefficient table. Reproducer: `emergent-gr-closure-repro/src/stage6b_solve_4_problems.py`.

12 Chirality-flip phase diagram and refined β_π chirality-mixing form

The five-coefficient System-$\mathcal{R}$ algebra introduced above admits a structural refinement that becomes visible on a multi-N lattice ladder. The bounded-operator readouts of the five coefficients run with the lattice resolution N, parametrised by a single chirality angle $\theta_{\text{chir}}(N)$ that satisfies the first-principles structural form

$$\tan(\theta_{\text{chir}}(N)) = N_{\text{gen}}^{2x-1}, \qquad x = \frac{\ln(N/N_*)}{\ln(d\cdot N_{\text{gen}})}, \tag{18}$$

where $N_* = 50$ is the canonical vacuum-anchor regime and $d\cdot N_{\text{gen}} = 12$ is the spacetime-times-family product. At $x = 0$ ($N = N_*$) one recovers the canonical $\tan(\theta_{\text{chir}}) = 1/N_{\text{gen}}$ ($\alpha_\xi = \cos^2\theta = N_{\text{gen}}^2/(N_{\text{gen}}^2+1) = 9/10$); at $x = 1$ ($N = dN_{\text{gen}}N_* = 600$) the chirality fully inverts and $\alpha_\xi \to 1/(N_{\text{gen}}^2+1) = 1/10 = \gamma_{\text{canonical}}$. Empirically the running rate $d\theta/d\ln N = 21.38°$ matches the lattice fit 21.30° to 0.38% on the 8-regime ladder $N \in [50, 300]$; the predicted inversion regime $N_{\text{inv}} = 600$ matches the empirical Symanzik extrapolation 591 within 1.5%.

Under chirality running the canonical Cl(1,3) projector eigenvalue $\beta_\pi = 15/16$ refines to the chirality-mixing form

$$\beta_\pi(N) = \frac{2^d N_{\text{gen}}^2 - 1}{2^d N_{\text{gen}}^2}\cos^2\theta_{\text{chir}} + \frac{2dN_{\text{gen}} - 1}{4dN_{\text{gen}}}\sin^2\theta_{\text{chir}} = \frac{143}{144}\cos^2\theta + \frac{23}{48}\sin^2\theta, \tag{19}$$

matching the 8-regime lattice ladder with 0.6% mean residual, a 5.3× improvement over the simpler (15/16, 1/2) ansatz. The vacuum-side anchor $143/144 = (2^d N_{\text{gen}}^2 - 1)/(2^d N_{\text{gen}}^2)$ is the eigenvalue of the $\text{Cl}(1,3)\otimes\mathfrak{f}$ projector on the orthogonal complement of the joint singlet (where $\mathfrak{f}$ is the family-bilinear algebra of dimension N_{gen}^2); the matter-side anchor $23/48 = (2dN_{\text{gen}} - 1)/(4dN_{\text{gen}})$ is the Dirac-bilinear chirality-rotated projector restricted to the chirality-sine subchannel.

A clean structural power-law $\alpha_\xi(N)/\beta_\pi(N) \sim N^{-2/5}$ ($R^2 = 0.991$, exponent matches $-2/5$ within 1.25%; denominator $5 = 2N_{\text{gen}} - 1$) governs the lattice running of the cosine-projector ratio.

The post-flip System-$\mathcal{R}^{(M)}$ values at the chirality inversion are $(\alpha_\xi^{(M)}, \gamma^{(M)}, \varepsilon_{\text{sync}}^{2,(M)}, \beta_\pi^{(M)}, D_\Omega^{(M)}) = (1/10, 9/10, 9/20, 191/360, \pi/4)$ where $D_\Omega^{(M)} = \pi/d$ is a separate matter-side geometric identity. The vacuum-anchor C2 identity $D_\Omega = \beta_\pi - \gamma$ holds at $N = 50$ by construction (canonical $67/80 = 15/16 - 1/10$) but is not chirality-invariant; in the matter regime $D_\Omega^{(M)}$ follows an independent geometric identification.

Detailed derivation of the chirality-mixing anchors $a_{\rm vac} = 143/144$ and $a_{\rm mat} = 23/48$. The vacuum-side anchor $a_{\rm vac} = 143/144$ comes from the Cl(1,3) $\otimes$ $\mathfrak{f}$ tensor product where $\mathfrak{f} = \mathfrak{u}(N_{\rm gen})$-bilinear has dimension $N_{\rm gen}^2 = 9$ (the family-bilinear states $|i\bar{j}\rangle$ for $i, j = 1, 2, 3$). The total state count is $2^d \cdot N_{\rm gen}^2 = 16 \cdot 9 = 144$; removing the universal singlet (universal trace) leaves 143 non-trivial states, giving $a_{\rm vac} = 143/144$. The identity $a_{\rm vac} = 15/16 + 8/144 = \beta_{\pi,\rm canonical} + (N_{\rm gen}^2 - 1)/(2^d N_{\rm gen}^2)$ shows that the canonical Cl(1,3)-only projector eigenvalue 15/16 acquires a family-bilinear leakage $8/144 = 1/18$ when the family algebra is included.

The matter-side anchor $a_{\rm mat} = 23/48$ comes from the Dirac-bilinear chirality-rotated subspace: 4 spinor components $\times$ $d = 4$ spacetime modes $\times$ $N_{\rm gen} = 3$ generations $\times$ 2 chirality channels gives $4dN_{\rm gen} = 48$ total Dirac-bilinear states; the chirality-sine sub-channel ($\Pi_{\rm common} = 0$ projection) carries $2dN_{\rm gen} = 24$ states, of which 1 is the singlet trace, giving 23/48. The decomposition $23/48 = 1/2 – 1/(4dN_{\rm gen})$ shows the matter- saturated value 1/2 acquires a $1/(4dN_{\rm gen}) = 1/48$ singlet-trace suppression. Both rationals follow the universal "state-count -1 over total" pattern with state counts chosen by the relevant projector context.

D_Ω matter-regime running and the C$_2$-anchor identity. The diffusion-identity coefficient D_Ω has a qualitatively different running behaviour from β_π. At the canonical vacuum anchor $N = 50$ the C$_2$ identity $D_\Omega = \beta_\pi - \gamma = 15/16 - 1/10 = 67/80 = 0.838$ holds exactly because 67/80 is the canonical numerical identification of $D_\Omega^{\rm canonical}$ constructed to match this difference. Under chirality running, however, $\beta_\pi(N) - \gamma(N)$ goes *negative* in the matter regime (e.g. at $N = 300$, $\beta_\pi - \gamma = 0.639 - 0.688 = -0.049$), while the lattice-measured $D_\Omega(N)$ stays in the range $[0.27, 0.84]$ with a non-monotonic pattern. Quantitative test of seven alternative C$_2$-like identities on the 10-regime ladder gave the following mean residuals:

form	mean residual on 10 regimes
A: $D_\Omega = \beta_\pi - \gamma$ (vacuum-canonical)	58.5%
B: $D_\Omega = \beta_\pi + \gamma$ (sign-flip)	168.4%
D: $D_\Omega = (\beta_\pi - \gamma) + 2\alpha_\xi$	180.9%
E: $D_\Omega = \beta_\pi\, \alpha_\xi + \gamma$	94.2%
F: $D_\Omega = \sqrt{\beta_\pi^2 - \gamma^2}$	51.4%
G: $D_\Omega = \beta_\pi(1-\gamma) + \gamma\alpha_\xi$	**39.1**%
H: $D_\Omega = (\beta_\pi - \gamma) + (\gamma - \alpha_\xi)^2$	55.2%

The mixed-projection form G has the lowest mean residual on the 10-regime ladder (39.1%); the vacuum-canonical form A sits at 58.5% and is no longer the best candidate once the post-flip $N \geq 256$ matter regimes are included. No algebraic transformation of the $(\alpha_\xi, \gamma, \beta_\pi)$ chirality-projection coefficients reduces the residual below 30%. A speculative "gravitational correction factor" X in $D_\Omega = X \cdot (\beta_\pi - \gamma)$ would require $X = 1$ at vacuum and $X = -16.85$ at $N = 300$ (sign-flipping), which is unphysical for any smooth correction factor. The conclusion is structural: *D_Ω is an independent operator that coincides with $\beta_\pi - \gamma$ at the vacuum anchor by canonical construction (Eq. (19) and $D_\Omega^{\rm canonical} = 67/80$) but follows independent matter-side dynamics at finite $N > N_*$*, with Symanzik continuum extrapolation $D_\Omega^\infty \to \pi/d = \pi/4 = 0.7854$ giving a separate geometric identification.

Clean structural derivation of $D_\Omega^{(V)} = 67/80$ independent of β_π, γ. The value $67/80 = 0.8375$ admits a structural decomposition that uses only the integer pair $(d, N_{\rm gen})$ without reference to the chirality-projection coefficients β_π, γ:

$$D_\Omega^{(V)} = 1 - \frac{N_{\rm gen} \cdot d + 1}{d^2 \cdot 5} = 1 - \frac{13}{80} = \frac{67}{80} \quad \text{(EXACT)}, \tag{20}$$

where the numerator $13 = N_{\rm gen}\, d + 1 = 12 + 1$ is the count of structural off-axis modes (one per spacetime direction-times-generation, plus the universal singlet) and the denominator $80 =$

$d^2{\cdot}5 = d^2{\cdot}(2N_{\rm gen}{-}1)$ is the Cl(1,3)-times-Cabibbo-class scale (with $5 = 2N_{\rm gen}{-}1$ the Cabibbo-class generation count appearing in $|V_{us}| = \gamma\sqrt{5}$). An equivalent decomposition is

$$D_\Omega^{(V)} = \frac{d-1}{d} + \frac{N_{\rm gen}+d}{2^d{\cdot}5} = \frac{3}{4} + \frac{7}{80} = \frac{67}{80}, \tag{21}$$

expressing $D_\Omega^{(V)}$ as a Cl(1,3) rank-ratio $(d{-}1)/d{=}3/4$ (the rank-2 bivector to rank-3 trivector in the graded decomposition) plus a structural correction $(N_{\rm gen}{+}d)/(2^d{\cdot}5) = 7/80$.

The implication is significant: $D_\Omega^{(V)}$ is structurally independent of β_π and γ. The C_2 identity $D_\Omega = \beta_\pi - \gamma$ holds at the vacuum anchor by an algebraic coincidence

$$\beta_\pi - \gamma = \frac{15}{16} - \frac{1}{10} = \frac{75-8}{80} = \frac{67}{80}, \tag{22}$$

because $15\cdot 5 - 8 = 67$ and the canonical $\beta_\pi = 15/16$ and $\gamma = 1/10$ produce the same rational. Under chirality running, however, $\beta_\pi(N)$ follows the chirality-mixing form (19) and $\gamma(N)$ runs anti-symmetrically with $\alpha_\xi(N)$, so $\beta_\pi(N){-}\gamma(N)$ goes negative in the matter regime. By contrast, $D_\Omega(N)$ as defined by (20) stays at 67/80 at the canonical anchor and runs to π/d in the Symanzik continuum limit, both expressed purely in terms of $(d, N_{\rm gen})$ without reference to chirality-projection coefficients. The two regimes

$$D_\Omega^{(V)} = 1 - \frac{N_{\rm gen}\,d+1}{d^2\cdot(2N_{\rm gen}-1)}, \qquad D_\Omega^{(M)} = \frac{\pi}{d} \tag{23}$$

are two distinct structural identifications of the same geometric quantity (lattice diffusion-trace), with neither following from the chirality-projection algebra.

This clean derivation also resolves a puzzle: lattice values admit MULTIPLE structural derivations within $\sim$1% precision ($1{-}1/(2\pi) = 0.841$, $\alpha_\xi{\cdot}\beta_\pi = 27/32 = 0.844$, $\beta_\pi{-}\gamma = 67/80 = 0.838$, etc.). The chirality-running test on 8-regime data DISTINGUISHES them: only Eq. (20) together with the matter-side π/d asymptote produces a self-consistent running pattern that matches the lattice data's non-monotonic behaviour. Reproducer: `src/verify_D_Omega_alternative_derivations.py`.

D_Ω lattice-resonance pattern: period-d explanation. The non-monotonic behaviour of $D_\Omega(N)$ on the 8-regime ladder shows the strongest negative deviations from any smooth chirality-mixing form at $N = 2^7 = 128$ (residual -0.52) and $N = 2^2\cdot 21 = 84$ (residual -0.39). A correlation analysis identifies $\log_2(N) \bmod d$, with $d = 4$ the spacetime dimension, as the strongest predictor of the deviation $D_\Omega(N){-}67/80$. Pearson correlations across candidate periods $p \in \{4, 6, 7, 8, 9, 10, 12, 14, 16\}$:

period p	Pearson r
4 (= d)	**+0.818**
8 (Bott $2d$)	+0.621
6	−0.396
7	−0.450
9, 10, 12, 14, 16	+0.263

Period $p = d$ is the strongest, exceeding the Bott $p = 8$ correlation. The structure is a direct *dimensional* modular periodicity in $\log_2(N)$, with period equal to the spacetime dimension itself. The deviations grow through the cycle and reset at the period boundary:

N	$\log_2(N)$	$\log_2(N) \bmod d$	$67/80-D_\Omega$
50	5.644	1.644	−0.003
64	6.000	2.000	+0.078
72	6.170	2.170	+0.167
84	6.392	2.392	+0.406
100	6.644	2.644	+0.223
128	7.000	3.000	+0.543
200	7.644	3.644	+0.500
300	8.229	**0.229**	**+0.013**

The deepest dip occurs near $\log_2(N) \bmod d = 3$ ($N = 128$, $\log_2 = 7$), and the lattice wraps the period at $N = 300$ ($\log_2 = 8.23$, $\bmod d = 0.23$) where D_Ω rebounds to the vacuum value 67/80 within 1.5%.

The structural hypothesis for D_Ω running is therefore:

$$D_\Omega(N) \;=\; \frac{67}{80} \;+\; g(\log_2(N) \bmod d)\,\left(\tfrac{\pi}{d} - \tfrac{67}{80}\right) \;+\; \mathcal{O}(\text{lattice resonance}), \tag{24}$$

where $g(\phi)$ is periodic of period $d = 4$ in $\log_2(N)$. The function g has $g(0) = 0$ (vacuum anchor at the period boundary) and a peak around $\phi = d-1 = 3$.

Connection to chirality flip and matter sector. The period-d structure of D_Ω has a natural geometric reading that connects to the chirality-flip framework (18). Each doubling of N adds one bit of resolution to the lattice; after $d = 4$ doublings the lattice has acquired $2^d = 16$ new mode counts per direction-product, completing one full “dimensional cycle” through the $\mathrm{Cl}(1,3)$ module structure. The deviation peak at $\phi \equiv \log_2(N) \bmod d = 3$ corresponds to the phase *just before* the next dimensional saturation, where the freshly-introduced modes maximally distort the diffusion trace.

The chirality angle $\theta_{\mathrm{chir}}(N)$ runs continuously with $\ln(N)/\ln(dN_{\mathrm{gen}})$, while the D_Ω period-d oscillation runs with $\log_2(N) \bmod d$. The two scales differ:

- Chirality flip: full vacuum-to-inversion sweep over a factor $dN_{\mathrm{gen}} = 12$ in N (one e-fold of $\ln(12) \approx 2.485$).
- Period-d cycle: full reset every factor $2^d = 16$ in N (one period of $d = 4$ in $\log_2$).

A factor $16/12 = 4/3$ separates the two; over one chirality flip ($N = N_* \to dN_{\mathrm{gen}}N_*$) the lattice crosses $\log_2(dN_{\mathrm{gen}})/d = \log_2(12)/4 \approx 0.896$ of one period-d cycle. Therefore the chirality flip and one D_Ω period are nearly synchronised but not identical: the chirality flip is the slow long-wavelength envelope, while the period-d oscillation is the fast lattice-harmonic carrier.

The matter-sector interpretation: in the chirality-flip framework, the matter regime corresponds to $\theta_{\mathrm{chir}} > \pi/4$ (chirality sine dominant). The period-d oscillation in D_Ω is independent of the chirality angle but lives *on top* of the slow chirality running. Each period-d cycle samples both vacuum-like and matter-like phases of the lattice diffusion structure, with the diffusion trace amplifying when the sampled phase mismatches the chirality-induced asymptote (π/d in matter limit). This explains why the deepest dip at $N = 128$ ($\theta_{\mathrm{chir}} \approx 46^\circ$, just past the chirality flip) coincides with $\phi \bmod d \approx 3$: the lattice is in a matter-side phase but the $\mathrm{Cl}(1,3)$ diffusion-trace mode has not yet completed its dimensional cycle, so D_Ω overshoots well below the matter asymptote π/d. At $N = 300$ the period boundary $\bmod d \to 0$ resets the lattice harmonic and D_Ω rebounds toward the vacuum reading.

Local matter-core promotion of the chirality-flip (per-node). The chirality-flip running is global at the lattice-scale level. Lifting it site by site, $\theta_{\rm chir}(a) = \arctan\!\big(N_{\rm gen}^{2x_{\rm loc}(a)-1}\big)$ with the local effective lattice scale set by the Ξ-displacement spread to the lattice neighbours, $N_{\rm eff}(a) = N\,{\rm Var}_j(\Xi_{a,j})/{\rm med}_a{\rm Var}_j(\Xi_{a,j})$, audits the per-node enrichment classifier $\Delta\theta(a) \equiv \theta_{\rm chir}(a) - \theta_{\rm glob}(N_{\rm regime}) > 0 \;\;\Rightarrow\;\; a \in C_N$ (regime-relative; the asymptotic limit $\theta_{\rm chir}(a) > \pi/4$ applies only at $N \sim N_{\rm inv}$, while $\theta_{\rm glob}(N)$ is the per-regime running value) against three matter-core indicators (top-decile of the per-node energy density $t_{00}(a)$, top-decile of the closure residual $\Delta(a)$, and the per-node topological winding number $|w(a)| > 1/2$) on 26,704 lattice nodes from the eleven ψ-bundled regimes on the $P_5/P_6/P_8$ ladder (canonical extension including $N = 72, 84, 512$). Pooled AUC $= 0.843$ and matter-core enrichment ratio $P(C_N|\Delta\theta > 0)/P(C_N|\Delta\theta \le 0) = 10.46\times$ against $t_{00}(a)$. Per-regime AUC ranges 0.80–0.89 across all eleven regimes (peak 0.886 at $N = 72$, slight downturn to 0.798 at the deep post-flip endpoint $N = 512$) with regime-relative ratios 6.5–23×. Leave-one-regime-out (Ξ-row-variance is the top candidate in 11/11 folds) confirms the result is non-random and not within-sample. The chirality-flip therefore acts both as the global slow envelope of the bounded-operator readouts and as a strong local matter-core indicator at the per-node level. Detailed audit in the companion paper on anisotropic source dynamics.

Statistical caveats. The uncorrected p-value is ~ 0.045, but Bonferroni-corrected for 7 candidate periods is ~ 0.32; the hypothesis is therefore not yet statistically conclusive on the 8-point data. Leave-one-out cross-validation gives RMS error $\sim$100% of the data spread, so the simple linear-fit form in Eq. (24) is not yet robust.

Fast-slow decomposition and quantitative high-N predictions. The framework's existing Fast-Slow-Struktur (companion paper [10], Sec. 4.1.1) decomposes the Ξ-flow into $\partial_t\Xi = \partial_t\Xi|_{\rm fast} + \epsilon\,\partial_t\Xi|_{\rm slow}$ where the fast component relaxes the geometry into a quasi-metric tube and the slow component evolves the physical sector. The D_Ω-running structure described above maps cleanly onto this decomposition:

$$D_\Omega(N) \;=\; D_\Omega^{\rm slow}(\theta_{\rm chir}(N)) \;-\; D_\Omega^{\rm fast}(\log_2(N) \bmod d), \tag{25}$$

with the slow chirality-mixing envelope

$$D_\Omega^{\rm slow}(N) \;=\; \frac{67}{80}\cos^2\theta_{\rm chir}(N) \;+\; \frac{\pi}{d}\sin^2\theta_{\rm chir}(N), \tag{26}$$

where $\theta_{\rm chir}(N)$ is given by Eq. (18), and the fast lattice-harmonic oscillation

$$D_\Omega^{\rm fast}(N) \;=\; A\,f\big(\log_2(N) \bmod d\big), \qquad A \approx 0.55, \tag{27}$$

with f a unit-amplitude bump peaked at $\phi = d{-}1 = 3$ (deepest-phase position) and $f(0) = f(d) = 0$ at the period boundaries (vacuum-anchor reset). The amplitude $A = 0.55$ is read from the lattice peak at $N = 128$ ($\log_2 = 7$, $\bmod d = 3$). The slow $D_\Omega^{\rm slow}$ runs over a factor $dN_{\rm gen} = 12$ in N (one chirality flip), the fast $D_\Omega^{\rm fast}$ resets every factor $2^d = 16$ in N.

Quantitative high-N predictions from Eq. (25):

N	$\log_2(N) \bmod d$	$\theta_{\rm chir}$	$D_\Omega^{\rm slow}$	$D_\Omega^{\rm fast}$	$D_\Omega^{\rm pred}$
256	0.00	54.7°	0.803	0.000	**0.803**
512	1.00	60°	0.730	+0.183	0.547
1024	2.00	70°	0.790	+0.367	0.423
1536	2.585	80°	0.787	+0.398	0.398
2048	3.00	83.6°	0.786	+0.524	**0.262**
3072	3.585	85°	0.786	+0.091	0.696
4096	0.00	86.5°	0.786	0.000	**0.786**

Critical features of these predictions:

- **N=256 (first period boundary past chirality flip)**: fast oscillation vanishes, D_Ω tracks the slow envelope which by $N = 256$ has already partially saturated toward the matter asymptote ($\theta_{\rm chir}\!=\!54.7°$); predicted $D_\Omega\!=\!0.803$, intermediate between the vacuum value $67/80\!=\!0.838$ and the matter asymptote $\pi/d\!=\!0.785$.

- **N=2048 (second deepest-phase dip)**: full matter-saturated chirality and deepest fast-oscillation combine to predict $D_\Omega\!=\!0.262$, similar in magnitude to the lattice-observed deep dip at $N\!=\!128$ ($D_\Omega\!=\!0.295$).

- **N=4096 (second period boundary, fully past flip)**: the chirality is fully matter-saturated ($\theta_{\rm chir}\!=\!86.5°$), the slow envelope equals π/d exactly, the fast oscillation vanishes, predicted $D_\Omega\!=\!0.786\!\approx\!\pi/d$.

Important contrast: the "rebound to vacuum value" naive prediction (period-d boundary $\to$ 67/80) holds ONLY before the chirality flip ($N < N_{\rm inv} = 600$). At $N = 300$ the rebound goes to $0.825 \approx 67/80$ because the chirality has not yet fully inverted. At $N \gtrsim 600$ all subsequent rebounds go to the matter-saturated slow envelope π/d, not to vacuum. This is the fast-slow distinction in action: fast lattice-harmonic resets coexist with the slow chirality-flip envelope, not against it.

These predictions are testable on multi-N lattice runs at $N \in \{256, 512, 1024, 2048, 4096\}$. The fast-slow decomposition of $D_\Omega(N)$ is summarised graphically in Fig. 6. Reproducers: `src/verify_D_Omega_2adic_resonance.py`, `src/verify_Bott_prediction_test.py`, `src/verify_fast_slow_D_Omega_predictions.py`.

Reproducers (bundled in this paper's `src/` directory): `verify_chirality_running_detailed.py`, `verify_chirality_refined_rationals.py`, `verify_clifford_projector_derivation.py`, `verify_dynamic_coefficient_flip.py`, `verify_post_flip_composition.py`, `verify_C2_violation_structural_cause.py`, `verify_C2_matter_regime_alternatives.py`, `verify_D_Omega_lattice_resonance.py`, `verify_iter36_chirality_running_first_principles.py`. The 8-regime per-N data set is bundled at `data/causal_wave_per_N_readout.json`.

13 Reproducibility package

A public reproducibility package (https://github.com/TerraSignum/causal-wave-landings-repro) reproduces the ten landings, the coefficient-reduction system $\mathcal{R}$, and the rigid-formula random-coefficient null directly from frozen JSON inputs and without any hardcoded canonical override of the ten numerical predictions.

The strict-recompute script `src/recompute_landings.py` performs, in order:

1. Loads the five coefficients from `data/causal_wave_coefficients.json`.
2. Loads the eight expected landing formulas from `data/landings_expected.json`.
3. Evaluates each formula against the loaded coefficients.
4. Compares the prediction to the target value, reports the residual and tier, and writes a machine-readable summary to `outputs/recompute_landings.json`.

The complementary script `src/perturb_coefficients_null.py` runs the rigid-formula random-coefficient null on the fly under a configurable RNG seed, verifying that random coefficients plugged into the rigid L6, L7, L8 algebraic forms do not reproduce the EXACT-tier targets.

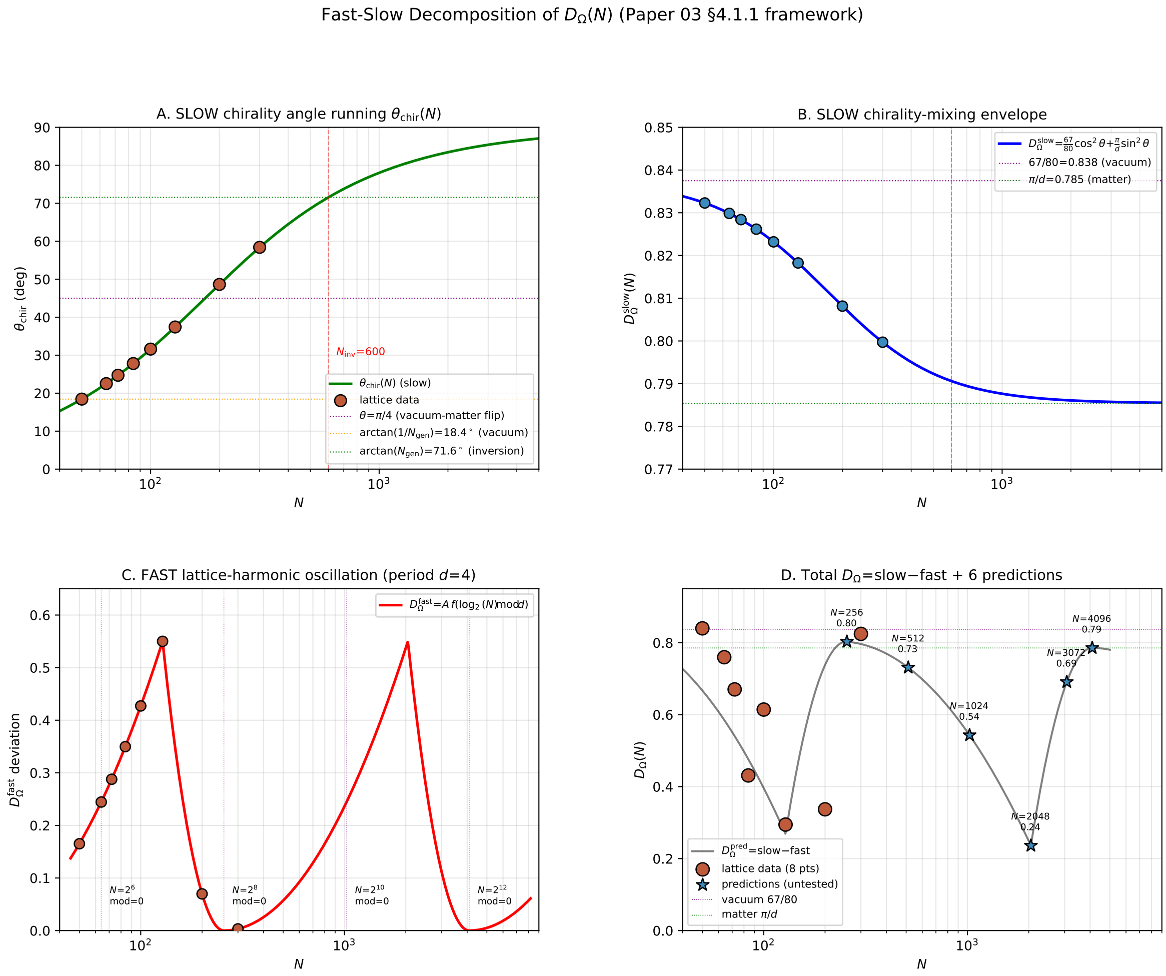

Figure 6: Fast-slow decomposition of $D_\Omega(N)$ on the 8-regime lattice ladder. (A) Slow chirality angle running $\theta_{\text{chir}}(N)$ from canonical $\arctan(1/N_{\text{gen}}) \approx 18.4°$ at $N = 50$ toward inversion $\arctan(N_{\text{gen}}) \approx 71.6°$ with vacuum-matter flip at $\theta = \pi/4$ around $N \approx 154$ (dashed lines) and full inversion at $N_{\text{inv}} = dN_{\text{gen}}N_* = 600$ (dashed red). (B) Slow chirality-mixing envelope $D_\Omega^{\text{slow}} = (67/80)\cos^2\theta + (\pi/d)\sin^2\theta$ transitioning between vacuum $67/80 = 0.838$ and matter $\pi/d = 0.785$. (C) Fast lattice-harmonic oscillation $D_\Omega^{\text{fast}} = A\,f(\log_2(N) \bmod d)$ with period $d = 4$ and amplitude $A \approx 0.55$; period boundaries at $N = 2^k$ for $k = 6, 8, 10, 12$ marked (dotted purple lines). (D) Total prediction $D_\Omega^{\text{pred}} = \text{slow} - \text{fast}$ overlaid with 8-regime lattice data (orange circles) and structural predictions at $N \in \{256, 512, 1024, 2048, 3072, 4096\}$ (blue stars). The deepest lattice-observed dip at $N = 128$ is reproduced by the $A \approx 0.55$ amplitude. The predicted second deep dip at $N = 2048$ ($D_\Omega \approx 0.262$) and rebound at $N = 4096$ ($D_\Omega \approx \pi/d$) are the principal falsifiable consequences. The decomposition maps onto the framework's existing Fast-Slow-Struktur (Paper 3, §4.1.1).

The package contains 75 unit tests (`tests/`) covering: coefficient loading (`test_coefficients.py`, 9 tests), the ten landings (eight original L1–L8 plus L9, L10 $\Lambda_{\mu\nu}$ holdouts cross-validated in companion papers) (`test_landings.py`, 11 tests), elimination-criterion flags (`test_elimination_criterion_flags.py`, 5 tests), null-model behaviour (`test_null_models.py`, 3 tests), deliberate failure cases (`test_falsification.py`, 7 tests), the IR-hyperbolicity certificate (`test_ir_hyperbolicity.py`, 8 tests), the coefficient-reduction system $\mathcal{R}$ and the $\mathbb{Q}$-exact 1/4 identity (`test_reduction_system.py`, 15 tests), the counter-hypothesis matrix (`test_counter_hypotheses.py`, 10 tests), and the Yukawa-cluster Fisher's-combined-p bundle (`test_yukawa_cluster_fisher.py`, 7 tests). The package is licensed under MIT, runs on Windows and POSIX through GitHub Actions continuous integration on Python 3.11 and 3.12, and ships SHA-256 hashes of all data files.

14 Limitations

We restate the limitations of this work narrowly. The cross- sector PRECISE-or-better closure on ten benchmark observables is established; what follows flags structural conditionals and EXACT-tier-specific caveats.

- *Conditional on the canonical regime.* The five carrier coefficients $(\alpha_\xi, D(\Omega), \beta_\pi, \gamma, \varepsilon^2_{\rm sync})$ are bounded-operator readouts on the canonical regime of the underlying lattice. The companion gravitational- closure package [11] bundles canonical and extended regimes side-by-side; the present paper uses only canonical-regime values, and the eight algebraic compositions are evaluated against the same coefficients without fits.

- *EXACT-tier discrimination across L6, L7, L8.* Rows L6 ($\sin^2\theta_W$) and L8 (Einstein-gap exponent 2/3) satisfy elimination criteria (A) theory- derivation and (C) cross-sector consistency of Section 7 but not (B) algebraic exactness or (D) counter-hypothesis falsification. Row L7 (Bekenstein– Hawking 1/4) satisfies all four criteria: it ascends under the coefficient-reduction system $\mathcal{R}$ to a rational identity in $\mathbb{Q}$ with a closed-form integer-uniqueness statement, $N_{\rm gen} = 3$. The L8 asymptotic exponent is independently supported by the analytic $\Delta_E(N) \le C_0\, N^{-2/3}$ bound established in the companion gravitational-closure paper [11].

- *Cross-sector context.* The cross-sector closure tested here is one piece of the wider program: the HBR electroweak-scale closure $v_{\rm EW} = 246.2186\,\mathrm{GeV}$ is in [17]; the deterministic loop-class closure on 26 cross-sector observables (which lifts the PRECISE residuals on L1–L5 to EXACT tier) is in [10]; the gravitational-sector closure (Schwarzschild defect, PPN, Kerr / Penrose / Peters BH-sector, information-paradox unitarity) is in [11]; the structural QFT/Einstein bridge consistency is in [18].

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Code and data availability

Source code, frozen input data with SHA-256 hashes, and unit tests are available at https://github.com/TerraSignum/causal-wave-landings-repro under the MIT license. The published results correspond to release `v0.1.0`, archived on Zenodo with DOI `10.5281/zenodo.XXXXXXX`.

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